

GEOLOGICAL SURVEY OF FINLAND
Ore Department

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**Mineral resources estimate for
the KEIVITSA Deposit**

Appendix to report: M06/3714/-94/1/10 Sodankylä, Keivitsa

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Summary

Detailed studies of the vast amount of drill core material from the Keivitsa Deposit have only just commenced. In this report, the size and grade of Keivitsa's mineral resources are outlined tentatively from one particular standpoint and using a simple method of estimation.

- The extent of the deposit in vertical and horizontal sections has been digitized using the analytical data for nickel alone. The volumes and tonnages of blocks corresponding to these digitized polygons have been estimated by the so called linear sectional method.
- Cutoff grade limits of 0.1 to 0.2 % Ni were chosen to delineate low grade types of mineralization, whereas grade limits of 0.2 to max % Ni were chosen for blocks of economic interest. This method may, however, be too simple for a multielement deposit like Keivitsa.
- Grades of the sum total estimate (Table 1.) are based on weighted averages counted on horizontal planes at 10 m intervals and simulating levels for mine planning. The grades for nickel, copper, gold, and platinum have all been calculated from the same samples chosen according to the above grade limits for nickel. Due to the technique used in compositing plane samples, the sum total grades include approximately 5 to 10 % dilution.

The combined mineral resources estimate is given in Table 1.

According to the estimate, **the total resources of the Keivitsa Deposit down to a depth of 300 meters are 112 Mt. The economically interesting portion amounts to 35 Mt.**

Table 1.

Combined estimate of the tentative planar mineral resources in the Keivitsa Deposit: probable and possible resources down to 100, 200, and 300 meters.

Cutoff % Ni	Depth m	Tonnage Mt		Ni %	Cu %	Au g/t	Pt g/t
		Probable	Possible				
0.1-0.2	100	18	2	0.14	0.16	0.06	0.16
	200	32	10	0.14	0.17	0.07	0.16
	300	52	25	0.14	0.17	0.06	0.16
0.2-max	100	8.6	1	0.32	0.36	0.13	0.47
	200	16	4	0.32	0.43	0.15	0.42
	300	24	11	0.32	0.44	0.15	0.4
0.1-max	300	76	36	0.19	0.25	0.09	0.24

Introduction, data, and evaluation procedures

The Keivitsa Prospect contains 287 drill holes in an area of 3.5 square kilometers, and the locations of drilling sites are shown in Figures 1 and 2. At its greatest, drilling density is nine holes per hectare. Information density is highest along the so-called Master Profiles, 3498900 (south-north, looking towards west) and 7512250 (west-east, looking towards north). In addition to the drill profiles, aligned parallel to the Master Profiles, modelling sections along diagonal directions, N1E - N10E (SW-NE, looking towards NW) and NWSE - NWSE2 (NW-SE, looking towards NE) were constructed. All these profiles and sections are shown in Figure 2.

Drill hole lengths vary between 9.55 m and 969.5 m and the total length drilled amounts to 37.5 km. Most of the holes are shorter than 60 m (198 holes) while a few are more than 400 m (26 holes). Correspondingly, hole density and, at the same time, sample density decreases overall with depth. There are about 300 samples at the surface (level +235 above the sea level), 70 samples at 105 m (+130 level), 45 samples at 235 m (0 level), 15 samples at 435 (-200 level), and only 2 samples at 635 m depth (-400 level).

Although several methods were used for element analyses, in this report it is assumed that the results are mutually comparable and reliable. The main emphasis has been on nickel and copper assays. In addition, cobalt, sulphur and noble element assays have been evaluated statistically as well as some noble elements for mineral resources estimates. Nickel, copper and cobalt assays amount to 16 070, sulphur assays to 14 738 while the noble element assays from original samples amount to less than 3 000. Therefore, calculations involving noble metals are less reliable than those involving Ni, Cu, Co and S. Noble elements were mainly analyzed later from combined samples amounting in total to 4 043 (see Appendix) and with combination intervals of 10 m. Original sample lengths vary from 0.1 to 6.4 m, the normal length being 2 m (11 201 samples), while 3 278 shorter samples and 1 591 samples that exceed this value.

A conventional means used for making the first mineral resources estimate for a deposit is the **linear cross section** method, which makes use of vertical drilling profiles with classified mineralized rock units outlined to form closed polygons. The areas occupied by each class in each section are then multiplied by section 'thickness' (distance of influence from a core sample) and density values (of rock types) to obtain volumes and tonnages. While each polygon has been assigned assay values calculated from drill core analyses, average metal contents and respective classifications can also be estimated. This method has been used in this study, with the following modifications:

- Instead of using drilling profiles only, **a network of WE- and SN- sections was constructed at 50 m intervals**. Polygons of mineralized rock types were interpolated along these sections, or extrapolated in cases with missing drill holes. In addition, **horizontal planes at 10 m intervals from +230 to +40 and at 50 m intervals from 0 to -400 were constructed** and polygons of mineralized rock types were outlined as above. **Resources were then estimated in all three directions, W-->E and S-->N and surface --> -400.**

- Instead of polygonwise assay averages **general class averages obtained by statistical analysis were used for mineral resources estimates** (Tables 1. and 6.). Averages were weighted by sample lengths. For the planar estimate (Tables 1 and 6b), samples were also composited according to horizontal planes, at 10 m intervals, from level +230 down to level -400. This

method is less accurate than the former, but in practice, due to the low and varying drilling density, differences in the final results are not significant and only mean that grade structure is not finely defined.

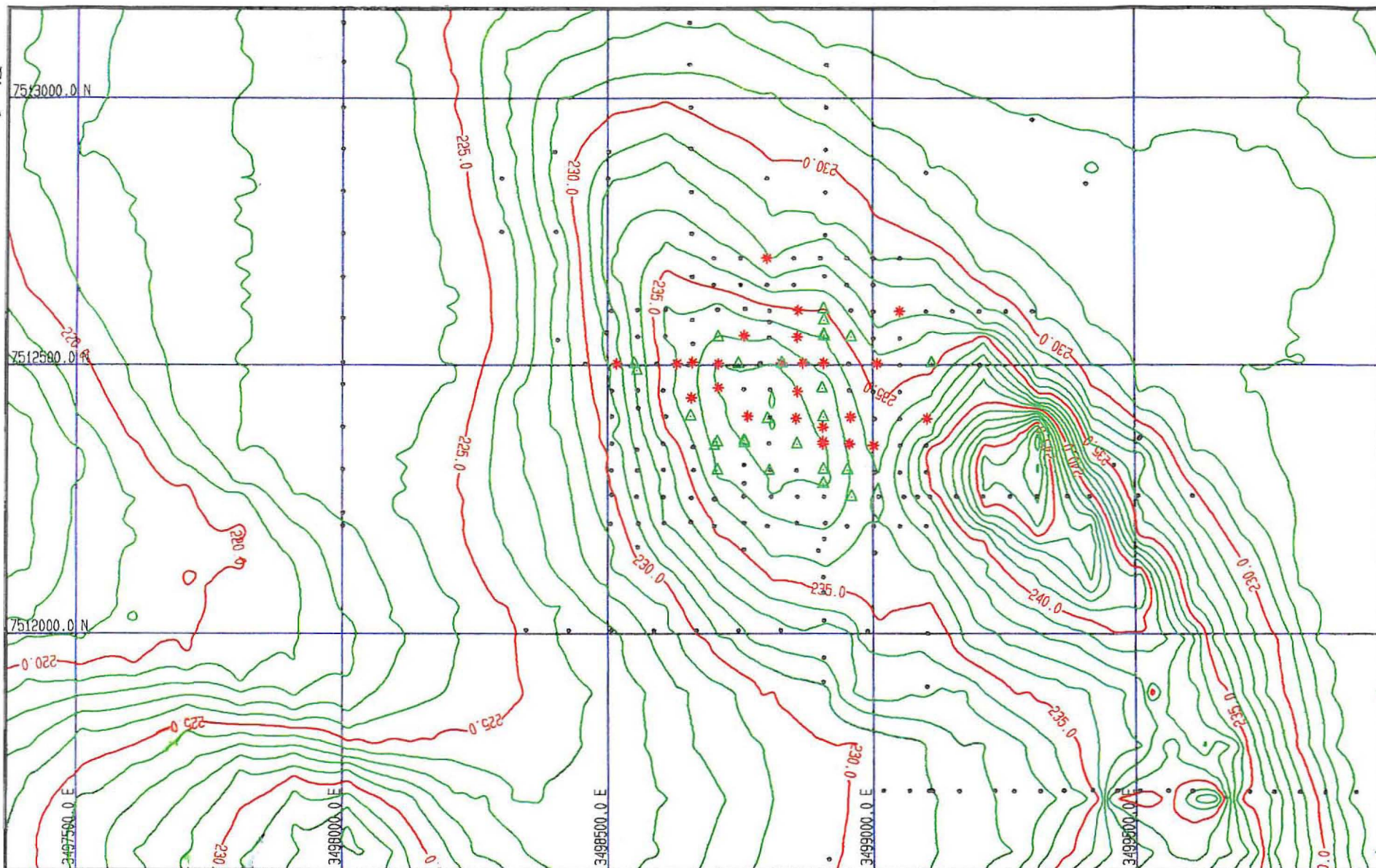
Mineralized rock types were defined according to cutoff assay values for nickel. The cutoff grades, 0.1, 0.2, and max % Ni, were based on the statistical behaviour of nickel. They also appeared when tested to be appropriate for outlining the shapes of mineralized bodies (Figures 4 - 9). Although the mineral resources estimate could have been based on mathematical interpolations, one of the aims of this study was to create a geological 3D model to describe and to study the structure of the deposit, allowing the possibility of extrapolating possible continuations. To create a perfect model, all available geodata would be necessary including, for instance, a petrographical classification of Keivitsa's rock types. The modelling and the resources estimate here are nevertheless based only on one element and on one method and, consequently, the results and conclusions must be treated with caution.

Since estimates are currently been made using other methods, the deposit will be ultimately described and evaluated more thoroughly and reliably than by using an one and only method. These other methods will probably include **block estimates** (interpolated even sized blocks, for example 10*10*10 m) or the **3D standard method** where polygons on sections are connected to corresponding polygons on other sections by triangulation of a net to cover uneven sized blocks. Block estimates will be necessary at least for mine planning, if not already at an earlier stage. 3D standard method modelling is currently in preparation at GSF.

Cutoff grades can also be **equivalence values** for instance in Ni or Cu. In the equivalence formula, weighting parameters for the different valuables depend not only on their market prices but also on their recoveries and on the cost limits imposed by smelting. An accurate formula can therefore be calculated only when the composition of the concentrate and the buyer are known. When evaluating polymetal deposits like Keivitsa it is, however, convenient to use equivalences for screening, even at an early phase of investigation.

In vertical and horizontal sections the deposit has been outlined using cutoff grades for nickel. Polygons grading from 0.1 to 0.2 % Ni are green coloured and polygons grading higher than 0.2 % Ni are red coloured. In some sections polygons high in copper (but not in nickel) have been coloured blue and those high in iron sulphides (but not in Ni or Cu) have been coloured magenta.

Statistical analyses, geological modelling and a mineral resources estimate of the Keivitsa Deposit were done with the help of software produced by Gemcom Services Inc. Geodata are available as a PC-XPLOR data base and a GEO-MODEL data base or as ASCII tables and HPGL- or DXF graphical files.



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Software by GEMCOM Services Inc.

KEIVITSA
Surface topography, drill holes, first samples
Ni 0.1 - 0.2 % green triangles
Ni > 0.2 % red asterisks

Fig. 1.
Keivitsa: topography, drill holes, and Ni assay values of first core samples.

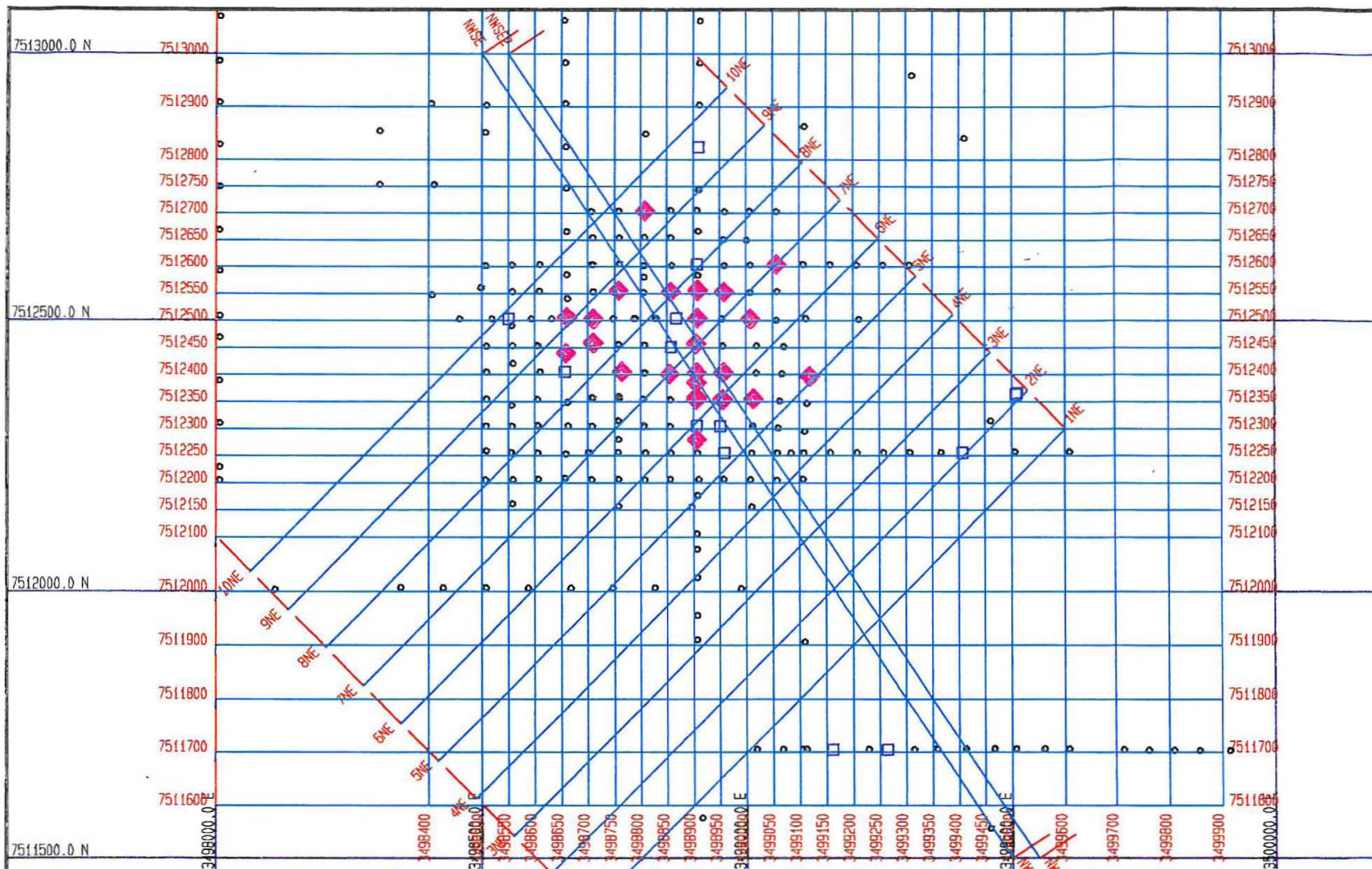


Fig. 2.
Keivitsa: drill hole profiles and modelling sections; Cu assay values of first core samples.

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KEIVITSA
Vertical profiles, drill holes, first samples

Cu 0.1 - 0.2 % blue squares

Cu > 0.2 % magenta diamonds

Statistical description of analytical data

Basic unweighted statistical data for different grade classes of nickel, copper, cobalt, sulphur, platinum, gold and rhenium are presented in Table 2. For example, the high maximum of 18.2 % for nickel is due to the assay value from a single 0.35 m long, rare and exceptional sample.

The above elements plus palladium and rhodium are described with logarithmic distribution diagrams and probability curves in Figure 3. These figures for the noble elements are based on 2 600 - 3 000 original non-combined sample analyses. It seems that, roughly speaking, all these elements exhibit a nearly lognormal distribution with two clear peaks in histograms. The curve forms of the probability plots confirm the conclusion that **there are at least two, possibly even four populations** present among the samples. This may result from the presence of several phases or several mixed layers and makes geological modelling difficult. In future studies, these populations should be defined and, if possible, they should be separated before further evaluations are made. In the case of nickel and copper, it seems that the **cutoff grade of 0.1 % roughly separates two populations**. The one with lower values may represent a 'normal' barren rock or host rock type, the petrography of which is so far unknown. The populations with assay values higher than 0.1 % Ni or Cu probably represent economically more interesting rock types than the former. **A grade of 0.1 % seems therefore suitable as the lowest cutoff grade limit for nickel and copper.**

Averages for six elements, weighted by sample lengths and based on cutoff grades of 0.1, 0.2 and 0.3 % for nickel and on cutoff grades of 0.1, 0.3 and 0.5 % for copper, are shown in Table 3 with total sample lengths of each grade class. Sample lengths for different grade classes of nickel, for instance, give a rough estimate of relative masses in the deposit: the total mass of the deposit will be reduced to about half when the cutoff grade for nickel is lifted from 0.1 to 0.2 %. Other dimensions may also depend on the length, and they will also be reduced accordingly; the mass reduction ratio should then be smaller than 0.5. **The average values in Table 3. are upper limits for expected assay values in a mineral resources estimate.** In reality it is not possible to define the mineralization precisely enough to avoid the problem of internal or marginal waste rock diluting mined rock.

Averages for four elements, weighted by sample lengths, composited according to horizontal planes at 10 m intervals from level +230 down to level -400, and classified according to cutoff grades as above, are given in Table 4. Each plane has a 5 m distance of influence both upwards and downwards. Composite samples created this way are 10 m long (steep holes) or in some cases longer (inclined holes). Here the resulting averages are lower than those in Table 3, since these composite samples necessarily include low grade samples as well. **These averages provide lower limits for expected assay values in a mineral resources estimate. Since they contain diluting material corresponding to internal or marginal waste rock they may approximate grades of mined rock.**

Correlation coefficients for a group of elements and for the coordinates X, Y, Z are given in Table 5A. The coordinate values were designated to check whether any grade/direction trends exist but none were detected. A significantly good correlation seems to hold between cobalt and sulphur.

A correlation matrix similar to Table 5A was calculated for a restricted part of the deposit, namely, the northeastern sector, starting from roughly the middle point at level +150. Here, mutual correlations between nickel, cobalt and copper are a little higher than those shown in Table 5A. All these elements also correlate negatively with the X coordinate values and less with the Y coordinate values: grades improve from the NE towards the SW. This may all imply that the deposit has a somehow circular form with a high grade zone in the middle. These results also support the use of statistically derived averages (Table 4.) for the mineral resources estimate. According to such results, some overestimation of grades may occur close to deposit margins and underestimation of grades may occur in the middle of the deposit. It may be significant that noble metals behave differently in that: no directional trend can be observed. Methodologically these results imply that difficulties will be accounted when trying to apply geostatistics to deposit evaluation; some difficulties may be solved with the help of noble metals.

In the calculations above for noble elements, original assay values for non-combined samples were used. Respective basic parameters and correlations have been calculated for 3984 combined samples (Appendix: Table 7).

The distribution of nickel and copper in space is depicted in Figures 4 -9. Grade values have been interpolated by using an inverse distance squared method with a 30 m search radius and without directional parameters. On horizontal planes the grid cells are 10*10 m in size and the scale is 1 : 10 000; on vertical sections the grid cell size is 5*5 m and the scale is 1 : 4 000. Interpolation probably defines the outer margin of the mineralization ($\text{Ni} > 0.1 \%$) quite sharply, but it exaggerates the area occupied by high grade mineralization ($\text{Ni} > 0.2 \%$).

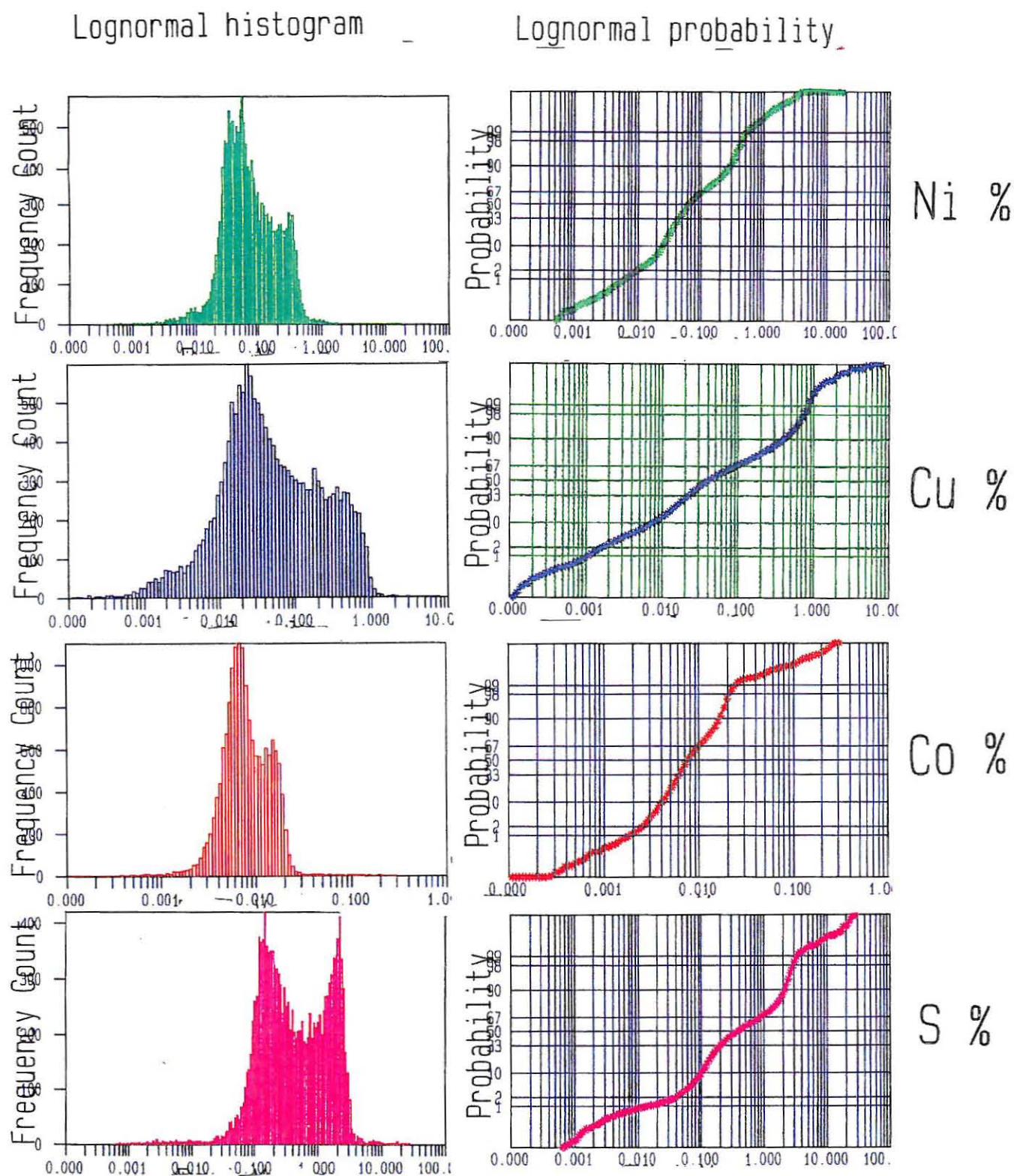


Fig. 3.

Keivitsa: distribution histograms and probability graphs of elements.

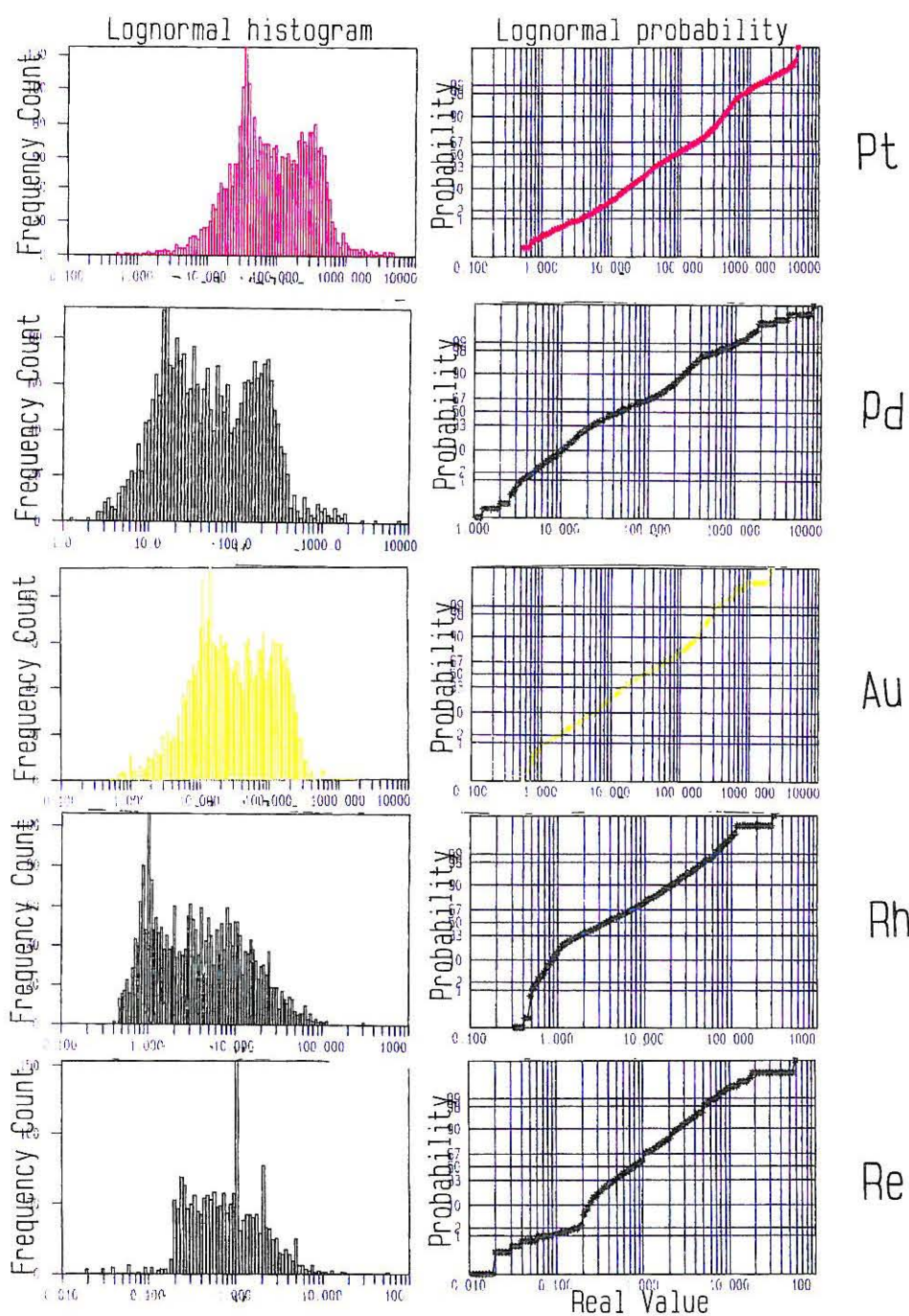


Fig. 3. (continuing).

Keivitsa: distribution histograms and probability graphs of elements.

Table 2.

Non-weighted grade averages classified according to different cutoff grades for each element.

Element	Samples	Min	Max	Average	Geom. mean	Median	Std.	CV	Skew	Kurtosis
Ni %	16070	0.0005	18.2	0.1169	0.07259	0.11433	0.20034	1.7138	48.03	4158.49
	5564	0.1	18.2	0.2483	0.2136	0.2314	0.29756	1.1982	40.95	2394.4
	2925	0.2	18.2	0.31478	0.31478	0.31526	0.38497	0.11875	35.23	1590.1
	1518	0.3	18.2	0.4356	0.3977	0.4044	0.51726	1.1876	27.46	920.5
	10506	0.0005	0.1	0.04728	0.0915	0.04098	0.02245	0.47472	0.39	2.426
	2639	0.1	0.2	0.14175	0.13898	0.1031	0.02826	0.19994	0.34	1.96
	1407	0.2	0.3	0.24629	0.2446	0.2447	0.0289	0.1175	0.145	1.82
	1013	0.3	0.4	0.34375	0.3426	0.3415	0.0278	0.0909	0.232	1.985
	323	0.4	0.5	0.4375	0.4366	0.433	0.0276	0.063	0.538	2.2
	74	0.5	0.6	0.53722	0.53649	0.53	0.0282	0.0525	0.521	2.14
Cu %	16061	0.0001	7.79	0.13268	0.04679	0.06124	0.21774	1.64106	6.47	144.24
	5122	0.1	7.79	0.35193	0.18462	0.27806	0.27725	0.78779	7.1	145.4
	3336	0.2	7.79	0.4628	0.41307	0.40318	0.2869	0.61995	8.83	179.96
	2372	0.3	7.79	0.55155	0.51273	0.49743	0.29693	0.53836	10.45	207.44
	10939	0.0001	0.1	0.03	0.0201	0.04174	0.02398	0.7985	1.072	3.374
	1786	0.1	0.2	0.1484	0.14194	0.14039	0.029	0.04074	0.194	1.81
	964	0.2	0.3	0.24442	0.2427	0.24417	0.02916	0.11929	0.214	1.819
	685	0.3	0.4	0.34712	0.346	0.34799	0.02791	0.0804	0.08	1.87
	528	0.4	0.5	0.44951	0.44858	0.42595	0.02891	0.06432	0.06	1.8
	387	0.5	0.6	0.54611	0.54534	0.5454	0.02915	0.05338	0.19	1.84
Co %	16067	0.0001	0.302	0.00927	0.00781	0.00777	0.0081	0.87429	13.66	343.17
	5164	0.01	0.302	0.01591	0.01485	0.01474	0.1147	0.72117	13.147	233.26
	458	0.02	0.302	0.03304	0.02728	0.02259	0.033	0.99868	4.54	27.3
	91	0.03	0.302	0.07682	0.06392	0.05328	0.0554	0.72121	2	6.83
	10903	0.0001	0.01	0.00613	0.00576	0.00619	0.00192	0.31339	-0.14	2.58
	4706	0.01	0.02	0.01424	0.01399	0.01428	0.00266	0.18714	0.17	2
	367	0.02	0.03	0.02219	0.02209	0.0217	0.00221	0.09952	1.56	4.99
S %	14738	0.0006	27.8	0.82315	0.41253	0.37602	1.113	1.35212	7.26	119.97
	8907	0.2	27.8	1.17678	0.8265	0.88652	1.21933	1.03617	7.5	115.9
	7172	0.4	27.8	1.50593	1.23784	1.33415	1.27617	0.84743	8.28	123
	5992	0.6	27.8	1.70559	1.48648	1.56148	1.30628	0.76588	8.88	128.5
	4520	1	27.8	2.00414	1.83346	1.30298	1.37658	0.68687	9.5	129.5
	1858	2	27.8	2.74714	2.54839	2.40384	1.88511	0.68621	7.81	76.3
	98	4	27.8	8.48541	7.23798	6.515	5.51514	0.64996	1.67	5.1
Pt ppb	2962	0.5	5000	182.1	84.4	85.2	311.2	1.7	7.3	85.3
	1332	100	5000	354.9	275.3	269.1	400.3	1.13	6.3	57.2
	899	200	5000	456.7	380.2	350.5	453	1	5.8	46.1
	569	300	5000	578.3	490.3	434.4	532.3	0.9	5	33.3
	192	500	5000	954	796.2	652	787.8	0.8	3.1	13.2
Au ppb	2781	0.5	2070	65.1	30.3	31.9	94.6	1.45	7.5	127.9
	645	100	2070	186.8	169.4	161.1	131.1	0.7	8.7	111.3
	184	200	2070	296.2	270.9	245.8	203.7	0.7	6.4	51.3
Rh ppb	2649	0.3	312	8.49	3.92	3.65	13.76	1.62	6.5	101.9
	1129	5	312	17.3	13.2	11.9	17.5	1	5.9	77.9
	671	10	312	24.2	20.2	18.3	19.9	0.82	5.8	69.7
	298	20	312	37.2	33.3	29.7	24	0.65	5.7	59.4

Table 3.

Weighted grade averages according to different cutoff grades for nickel and copper.

NI cutoff %	NI % length m	Cu % length m	Co % length m	S % length m	Au ppb length m	Pt ppb length m
0.1	0.24 10617	0.31 10617	0.014 10617	1.65 9162	95 5958	254 5985
0.2	0.33 5519	0.45 5519	0.016 5519	1.87 5473	136 3032	358 3101
0.3	0.41 2897	0.54 2897	0.014 2897	2.16 2893	172 1425	483 1456

Cu cutoff %

0.1	0.23 9907	0.34 9907	0.015 9907	1.61 9602	104 5549	224 5599
0.3	0.31 4573	0.54 4573	0.017 4573	2.01 4460	147 2504	289 2552
0.5	0.35 2276	0.7 2276	0.019 2276	2.27 2222	182 1013	331 1061

Table 4.

Weighted grade averages on horizontal planes according to different cutoff grades for nickel and copper.

NI cutoff %	NI %	Cu %	Co %	S %
0.1	0.19	0.25	0.012	1.2
0.2	0.31	0.43	0.015	1.7
0.3	0.4	0.53	0.018	2

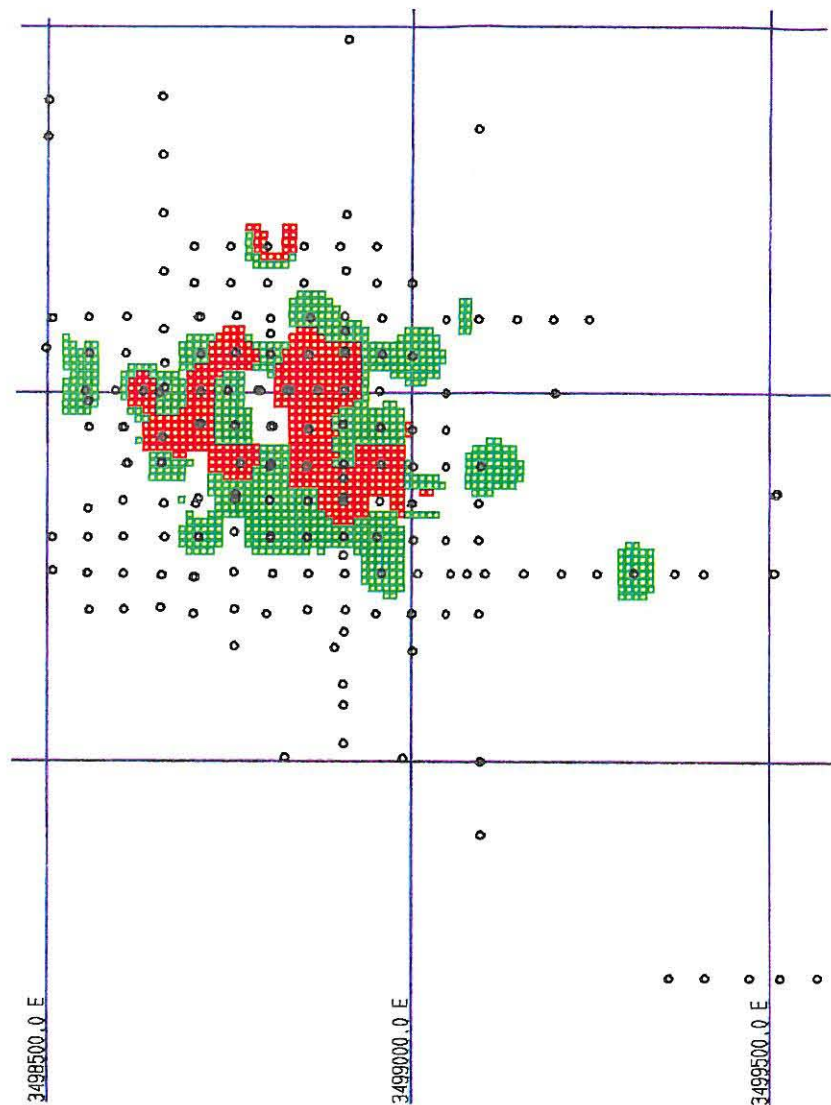
Cu cutoff %

0.1	0.21	0.32	0.013	1.4
0.3	0.3	0.53	0.016	1.9
0.5	0.35	0.7	0.019	2

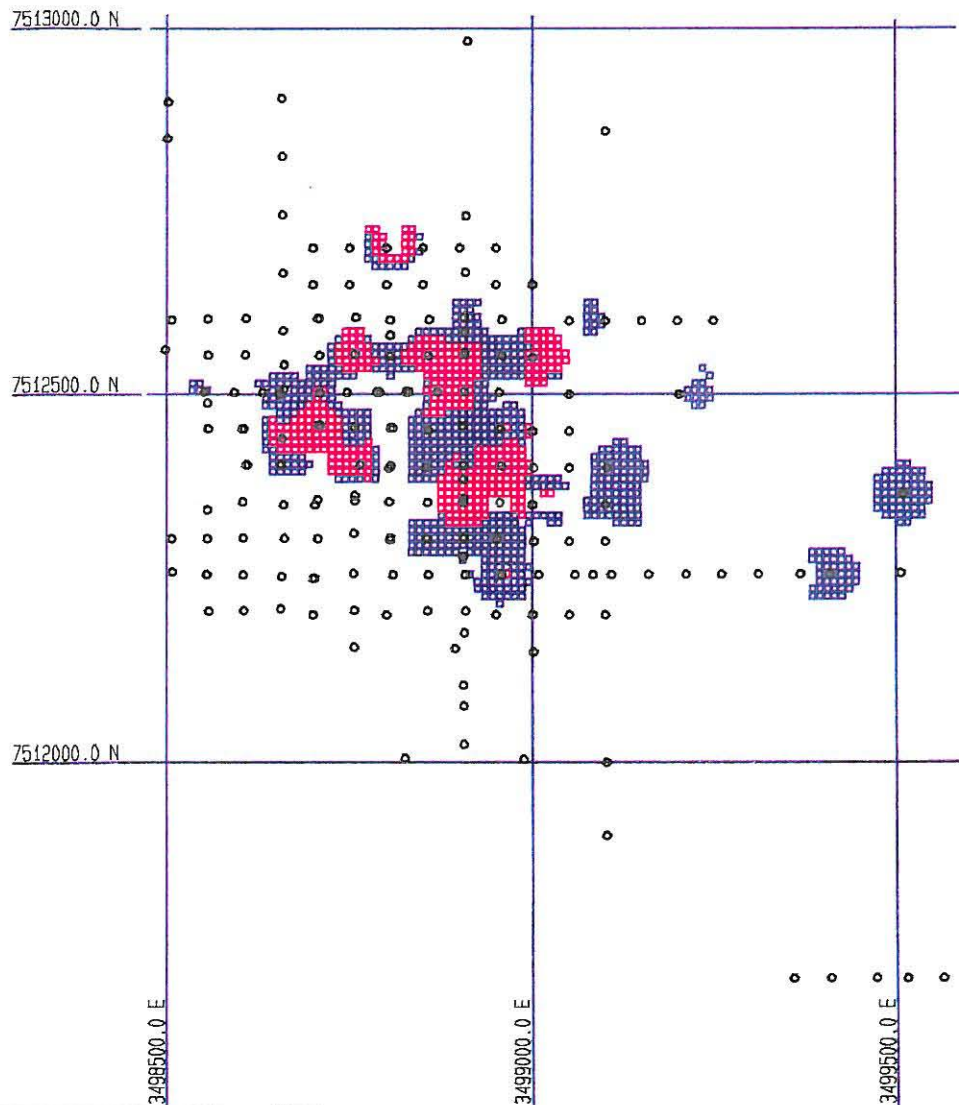
Table 5.

Correlation matrices: A. All assays. B. Assays from the NE quarter of the Deposit.

A.	CU	NI	CO	S	Au	RH	PD	RE	PT	X-COORD	Y-COORD
CU	-----										
NI	0.45367	-----									
CO	0.47158	0.54925	-----								
S	0.55386	0.54523	0.89221	-----							
AU	0.59488	0.43157	0.26334	0.31101	-----						
RH	0.56049	0.56347	0.19795	0.2569	0.50537	-----					
PD	0.1736	0.54646	0.09507	0.10324	0.30659	0.4668	-----				
RE	0.12796	0.09799	0.23721	0.32428	0.05512	0.04861	-0.00081	-----			
PT	0.22739	0.59064	0.07757	0.09644	0.36537	0.59048	0.86295	-0.02542	-----		
X-COORD	-0.0185	0.00834	0.01698	0.02615	0.02802	0.03469	0.07883	0.03673	0.08839	-----	
Y-COORD	0.05085	-0.00316	0.07531	0.06501	0.00046	-0.03342	-0.07745	0.03206	-0.08318	-0.31642	-----
Z-COORD	-0.11062	-0.06908	-0.08285	-0.1215	-0.00288	-0.04219	0.05967	-0.04658	0.06849	0.21187	-0.26416
B.	CU	NI	CO	S	Au	RH	PD	RE	PT	X-COORD	Y-COORD
CU	-----										
NI	0.79484	-----									
CO	0.5306	0.79432	-----								
S	0.57667	0.66152	0.82655	-----							
AU	0.18587	0.22115	0.16511	0.1386	-----						
RH	0.18562	0.31292	0.28091	0.17931	0.75861	-----					
PD	0.2181	0.44007	0.37492	0.22641	0.8218	0.84	-----				
RE	0.05319	0.18517	0.35646	0.29382	0.48447	0.5748	0.52106	-----			
PT	0.19749	0.28803	0.21026	0.15675	0.9046	0.83783	0.93525	0.5129	-----		
X-COORD	-0.35235	-0.32101	-0.20887	-0.34466	-0.07215	-0.05155	-0.0746	-0.01268	-0.07403	-----	
Y-COORD	-0.20112	-0.22431	-0.13202	-0.07652	-0.11881	-0.08118	-0.11316	-0.03682	-0.12289	-0.31642	-----
Z-COORD	-0.05732	-0.02053	-0.05773	-0.24196	0.09816	0.10297	0.10246	0.07188	0.12218	0.21187	-0.26416



KEVITSA
Planview + 230
Drill holes, Ni averages
Ni 0.1 - 0.2 % green, > 0.2 % red

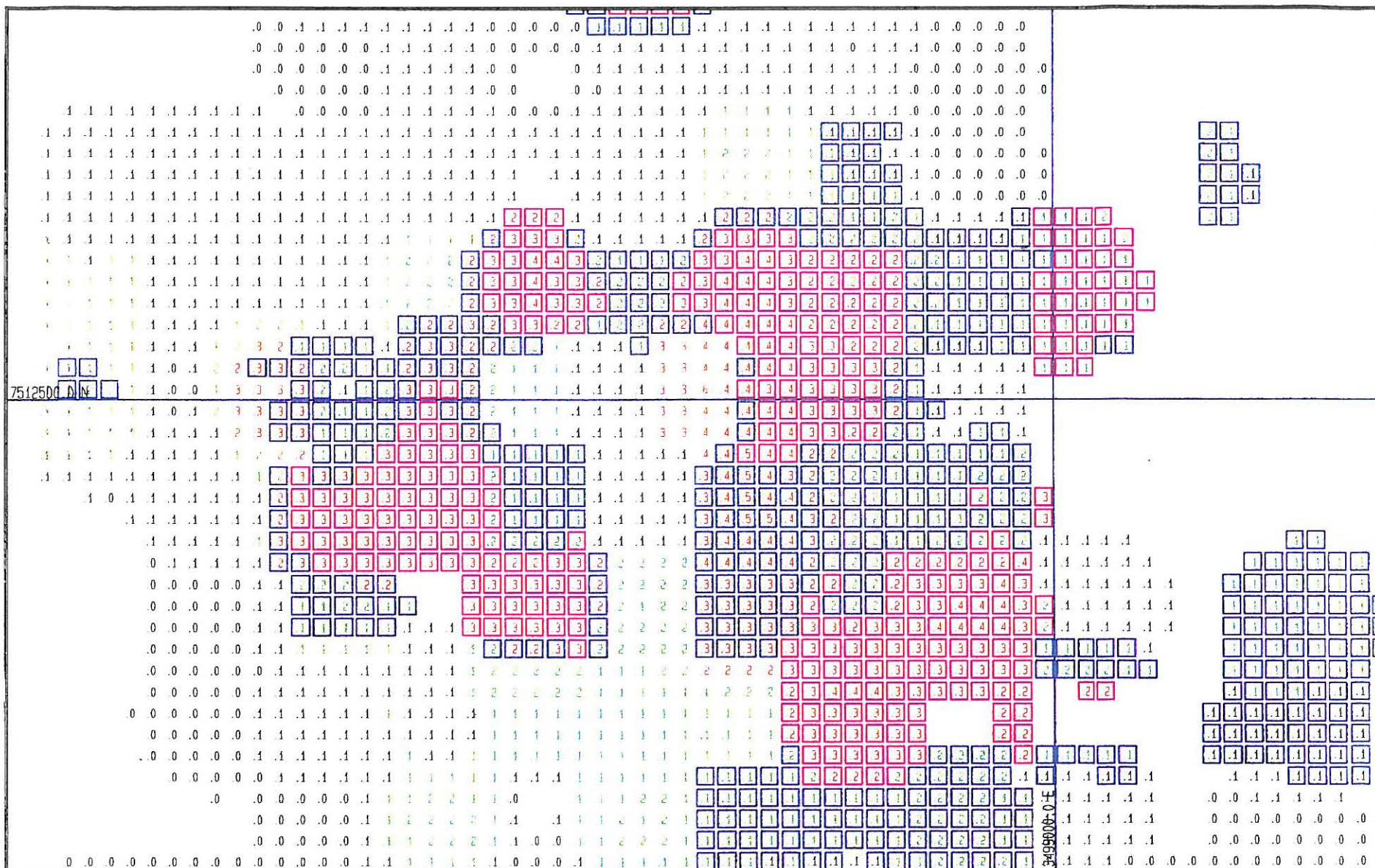


KEVITSA
PLANVIEW + 230
Drill holes, Cu averages:
Cu 0.1 - 0.3 % blue, > 0.3 % magenta

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Fig. 4.
Keivitsa, level +230, interpolations of nickel and copper grade values.



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KEVITSA
Planview + 230
Drill holes, Ni & Cu averages
Ni assay values. Cu blue-magenta squares

Fig. 5.
Keivitsa, level +230, a detail of data in Fig. 4.

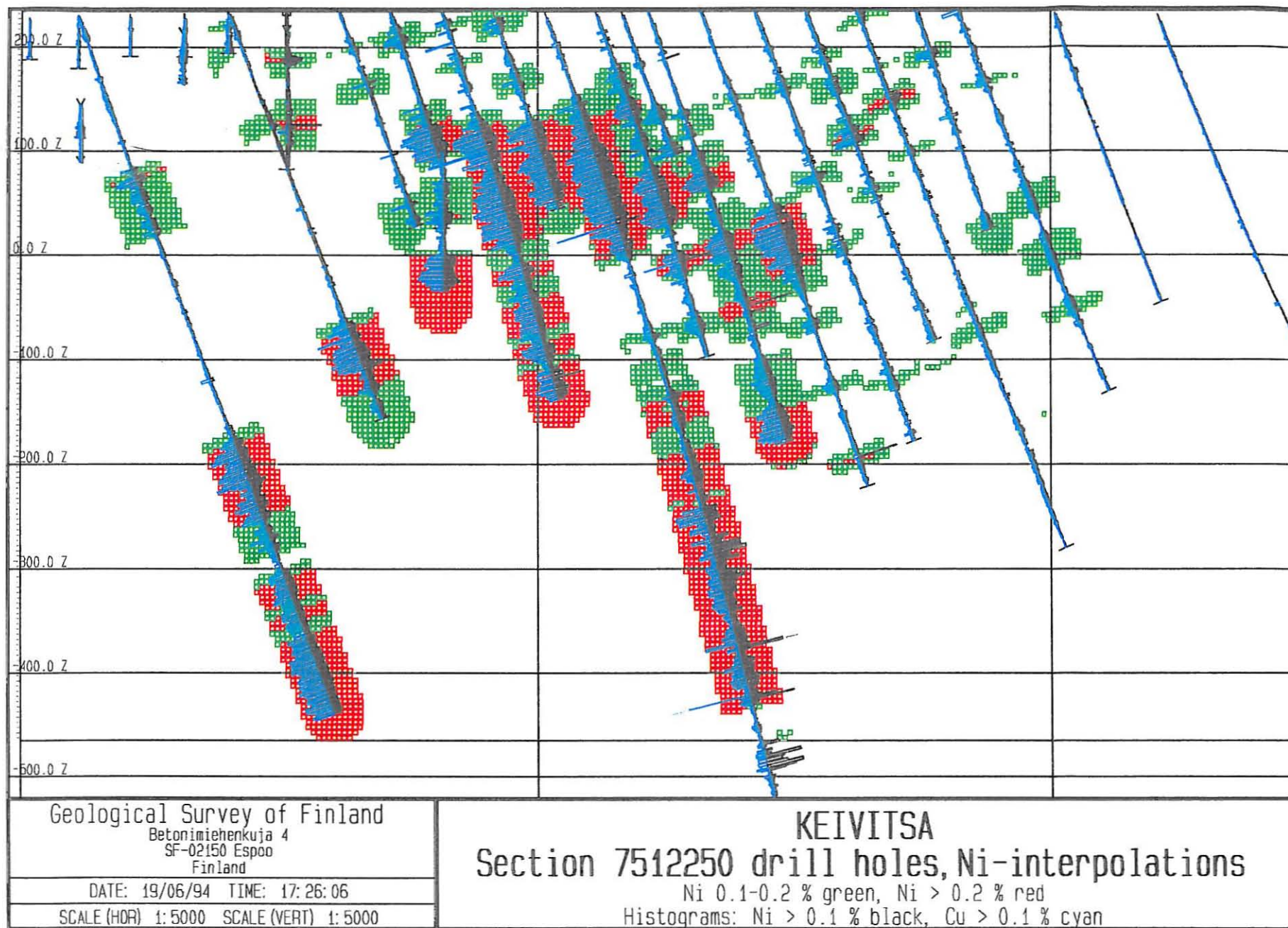


Fig. 6.
Keivisa, vertical section 512250, interpolation of nickel grade values (Master Profile).

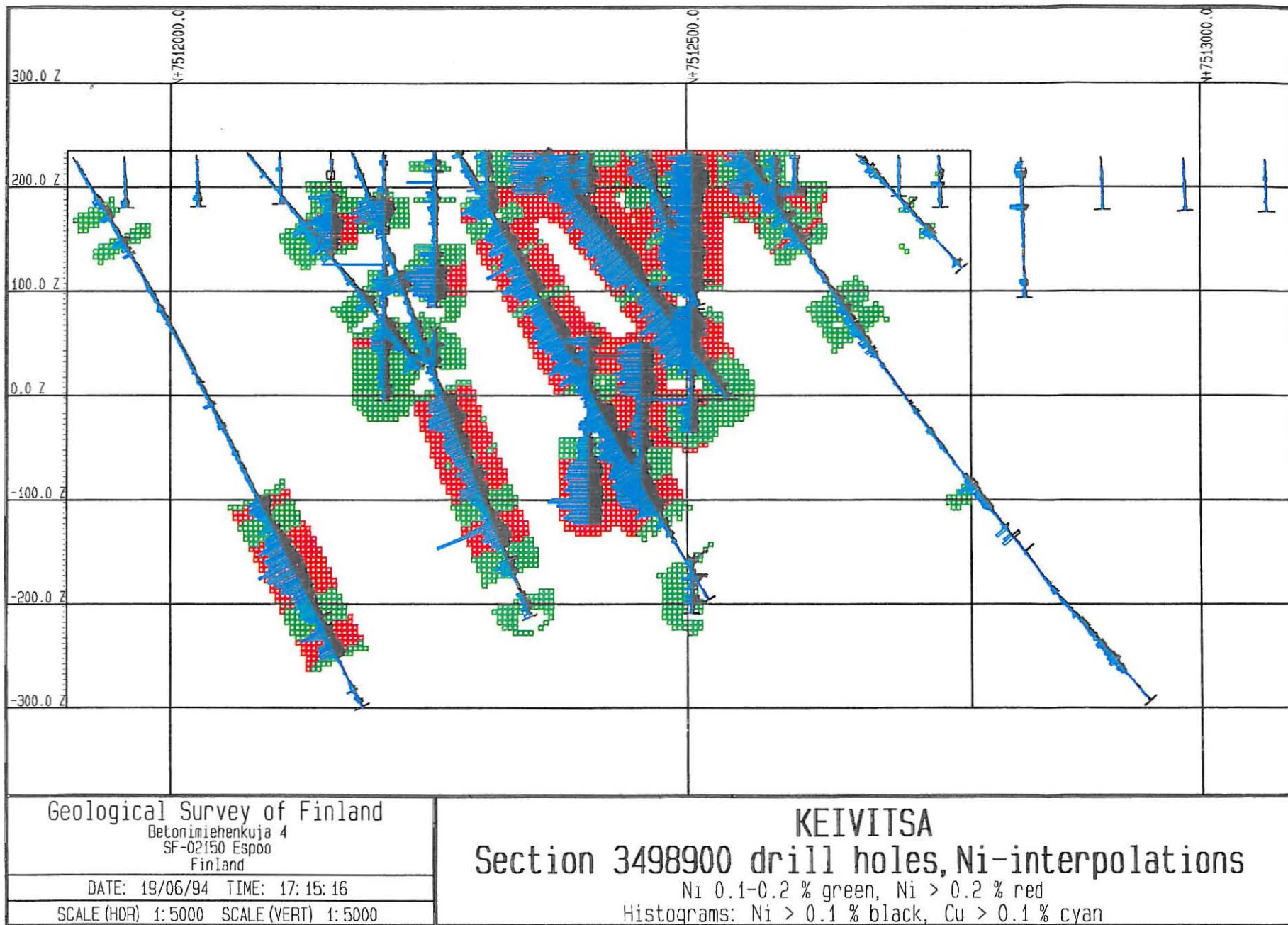


Fig. 7.
Keivitsa, vertical profile 3498900, interpolation of nickel grade values ('Master Profile').



Fig. 8.
Kelvissa, vertical section 7NE (SW-NE), interpolation of nickel grade values.

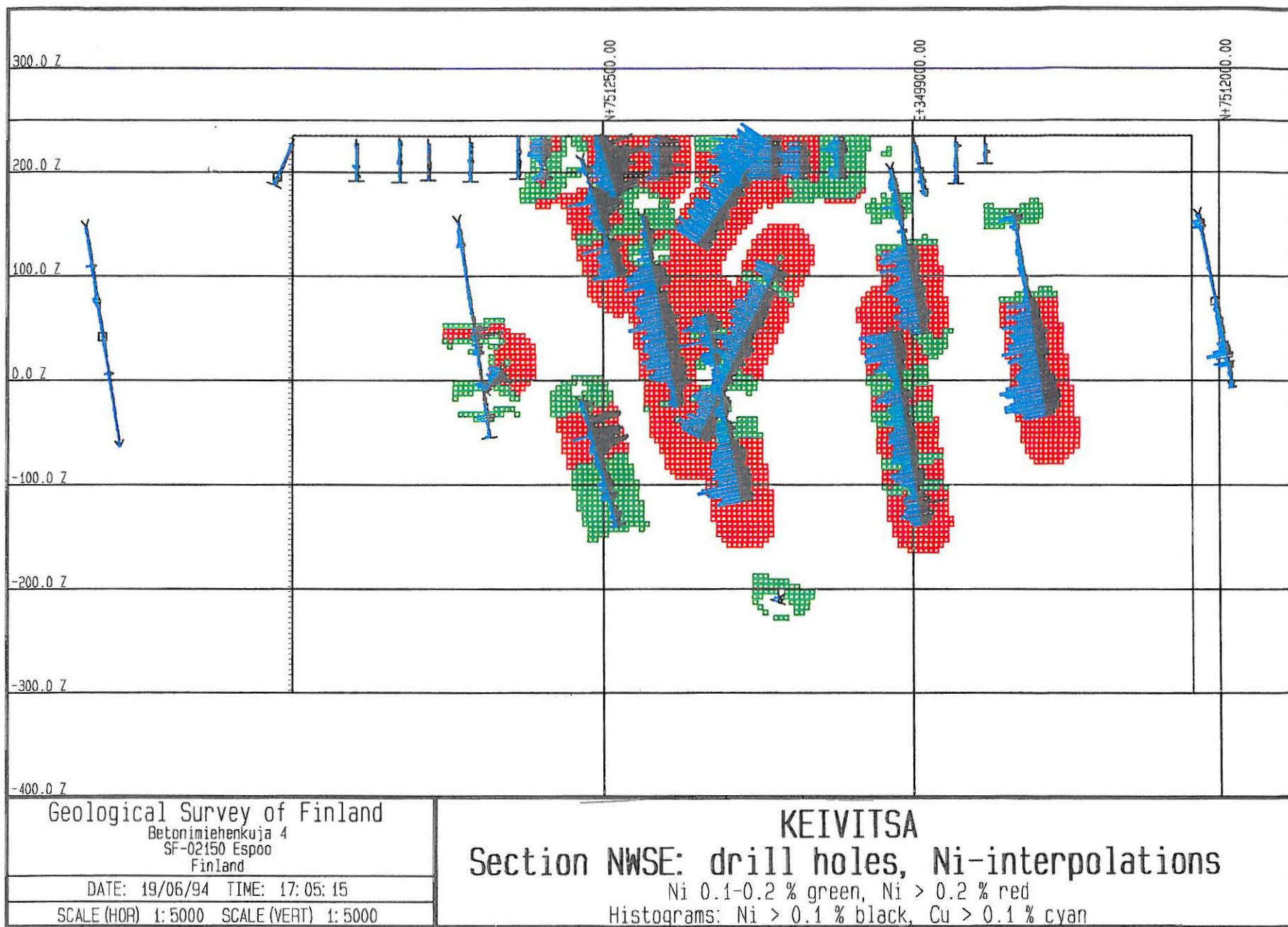


Fig. 9.
Keivitsa, vertical section NWSE (NW-SE), interpolation of nickel grade values.

Geological modelling of the Keivitsa Deposit

Geological deposit modelling aims at an analysis and description of geological and technical rock or ore units, and their relationships and structures, that are relevant for evaluation, further investigations, and mining development. In this tentative case, modelling is restricted to the grade structure of the Keivitsa Deposit and is based on nickel assays.

The method used here consisted of **iterative outlining of grade patterns** on vertical sections in different directions and on horizontal planes. The starting phase included statistical analysis of elements in order to find out their distribution parameters and suitable cutoff grade limits. It was followed by an inverse distance grade interpolation on the above sections (figures 4-9). Although the distributions for all elements appeared to exhibit lognormal symmetry, interpolations were made without transformation of grade values. It remains to be studied later whether this decision was significant or purely cosmetic (as I think now). The interpolation phase led to the digitizing of equal value areas with arising concepts of deposit structures. The first interpreted features appeared to vary considerably depending on sample density and section orientation. The next phase consisted of harmonizing these interpretations. This work was not carried very far, since harmonizing masks alternate interpretations (none of which are necessarily correct). The final phase would be to complete this harmonizing and to create a three-dimensional solid model of the deposit.

The **incomplete harmonizing** between corresponding patterns on sections in different directions **results in inequalities in mineral resources estimates** as is seen in Table 6. In part, these differences result from the varying number of sections used.

The nickel grade structure of the Keivitsa Deposit is nevertheless rather clear: the **mineralization has the form of an inverted boat**. The keel of the boat intersects the surface across an area corresponding to the southern margin of the outcropping deposit (near northing coordinate 7512300, see Figs. 15-18). The keel plunges down to south-southeast with a dip of 20 - 40 degrees parallel to the longitudinal or axial direction of the deposit. When continuing southwards along this axial trend, the deposit bifurcates into two branches corresponding to the hull of the boat. **The keel and the hull represent the most confidently delineated parts of the deposit.** West-east sections are the most convenient for resource estimation because they are almost perpendicular to this axial direction.

As can be judged from Figures 15 - 18, the above interpretation is not the only one possible. To point out a few weaknesses, many coloured polygons are extrapolations with little or no drill hole support, for example on sections 7511900 - 7512100, 7512200. Furthermore, is it actually the absence of drill holes in the middle of Figure 13, section 7512250, which results in the boat like morphology? Nevertheless, this model does constrain the minimum size of the deposit and helps identify areas where further drilling is necessary.

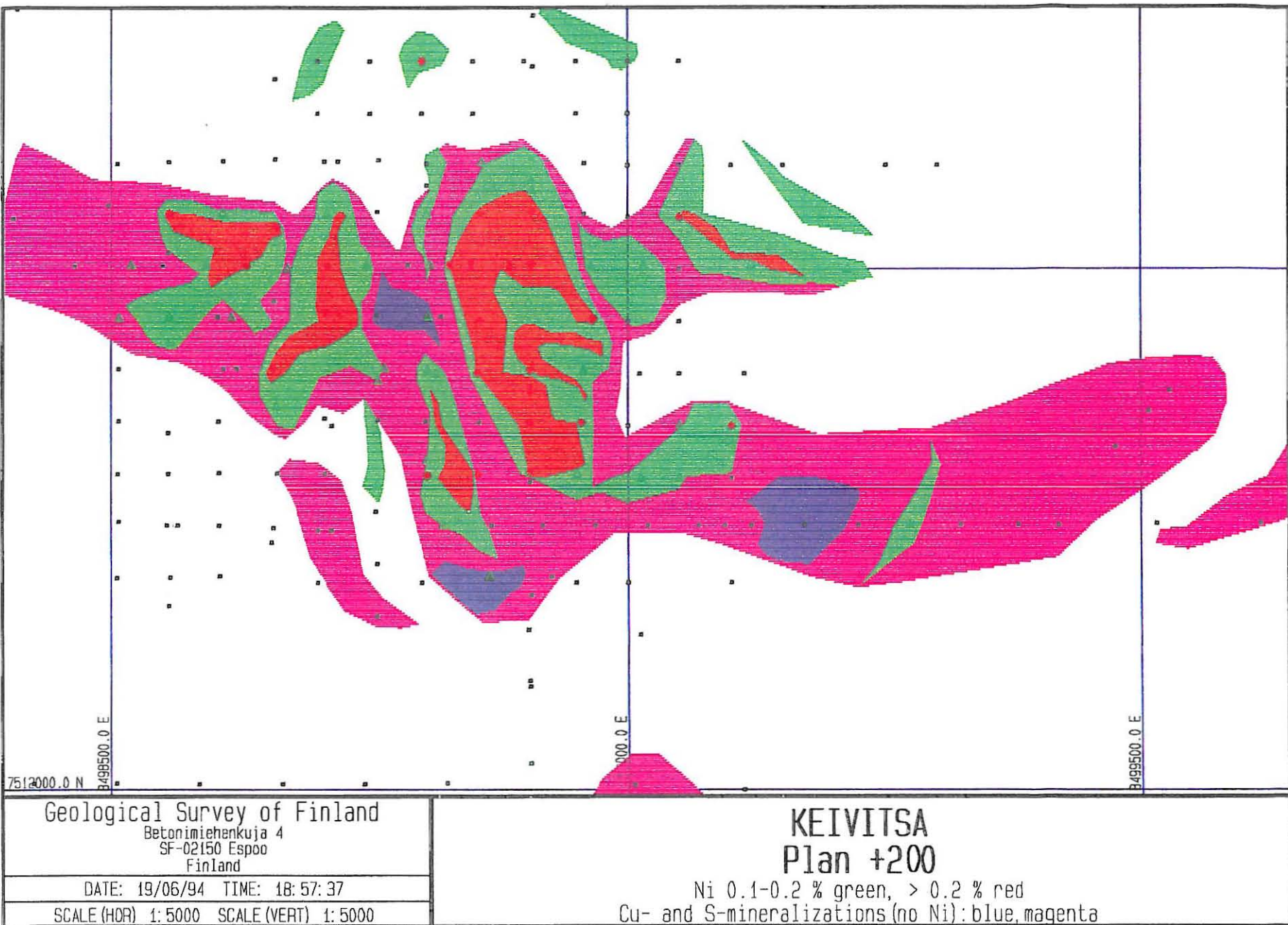


Fig. 11.
Keivitsa, level +200. Mineralizations: green = Ni 0.1 - 0.2 %, red = Ni > 0.2 %, blue = Cu mineralization (no Ni), magenta = iron sulphide mineralization (no Ni, Cu).

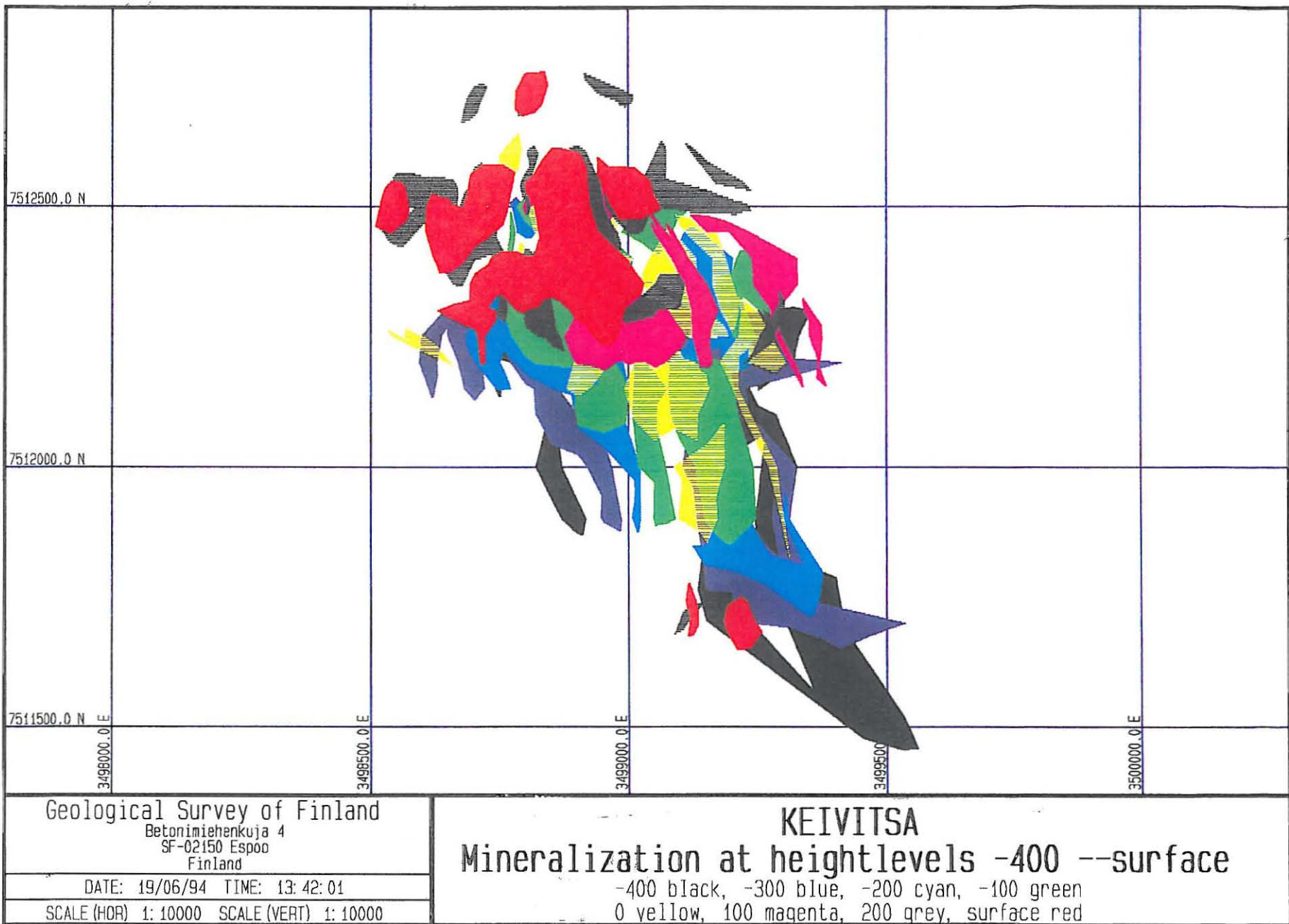
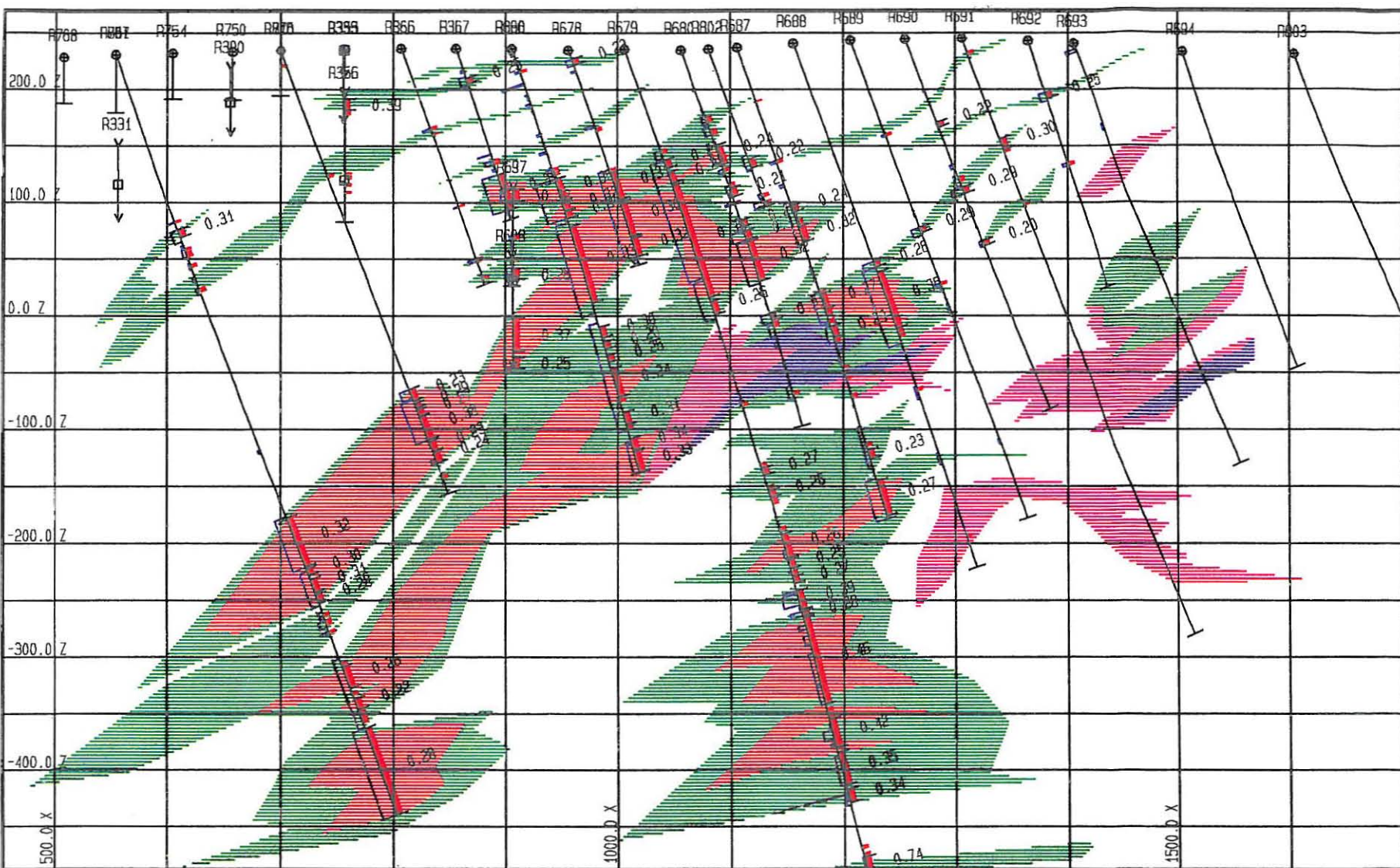


Fig. 12.

Keivitsa, Ni mineralization (Ni >0.1 %) outlined on horizontal planes from level -400 up to surface: -400, -300, -200, -100, 0, +100, +200, surface (+235) with colours in respective order: black, blue, cyan, green, yellow, magenta, grey, red.



Geological Survey of Finland
Betoniemiehenkuja 4
SF-02150 Espoo
Finland

DATE: 16/06/94 TIME: 15:44:54

SCALE (HOR) 1: 5000 SCALE (VERT) 1: 5000

Section = 7512250 Outline = Litholog
Keivitsansarvi

Ni-mineralisation, grades, cutoff 0.1 %
0.1-0.2% green, >0.2% red (Cu: blue histogr.)

Fig. 13.

Keivitsa, west-east Master Profile 7512250: Ni mineralization, green = 0.1 - 0.2 % Ni, red = > 0.2 % Ni; blue = Cu mineralization, magenta = iron sulphide mineralization. Higher class Ni grades of drill cores with red bars, and grade values, Cu with blue histograms.

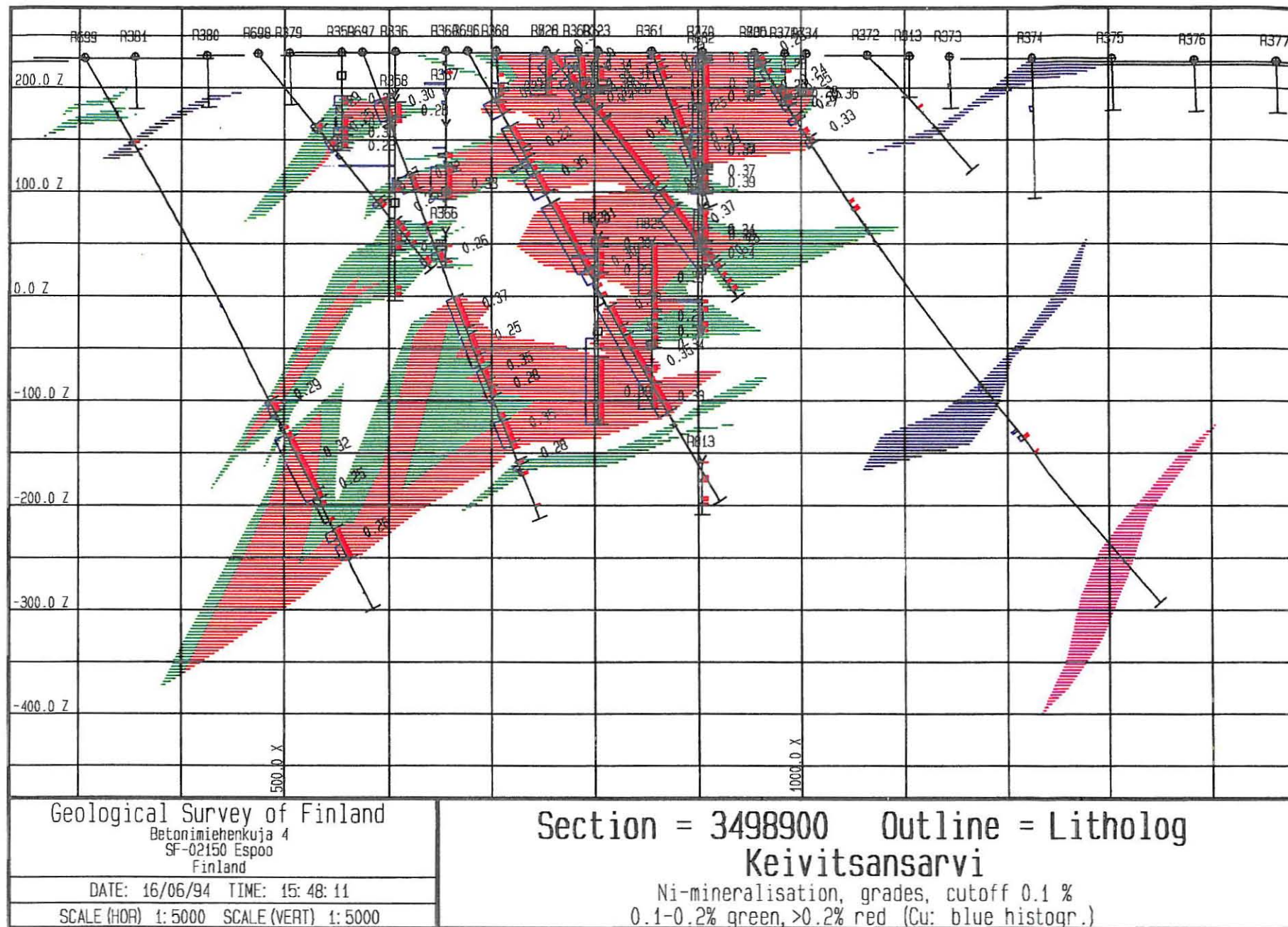


Fig. 14.
Keivitsa, south-north 'Master Profile' 3498900. Explanation as in Fig. 13.

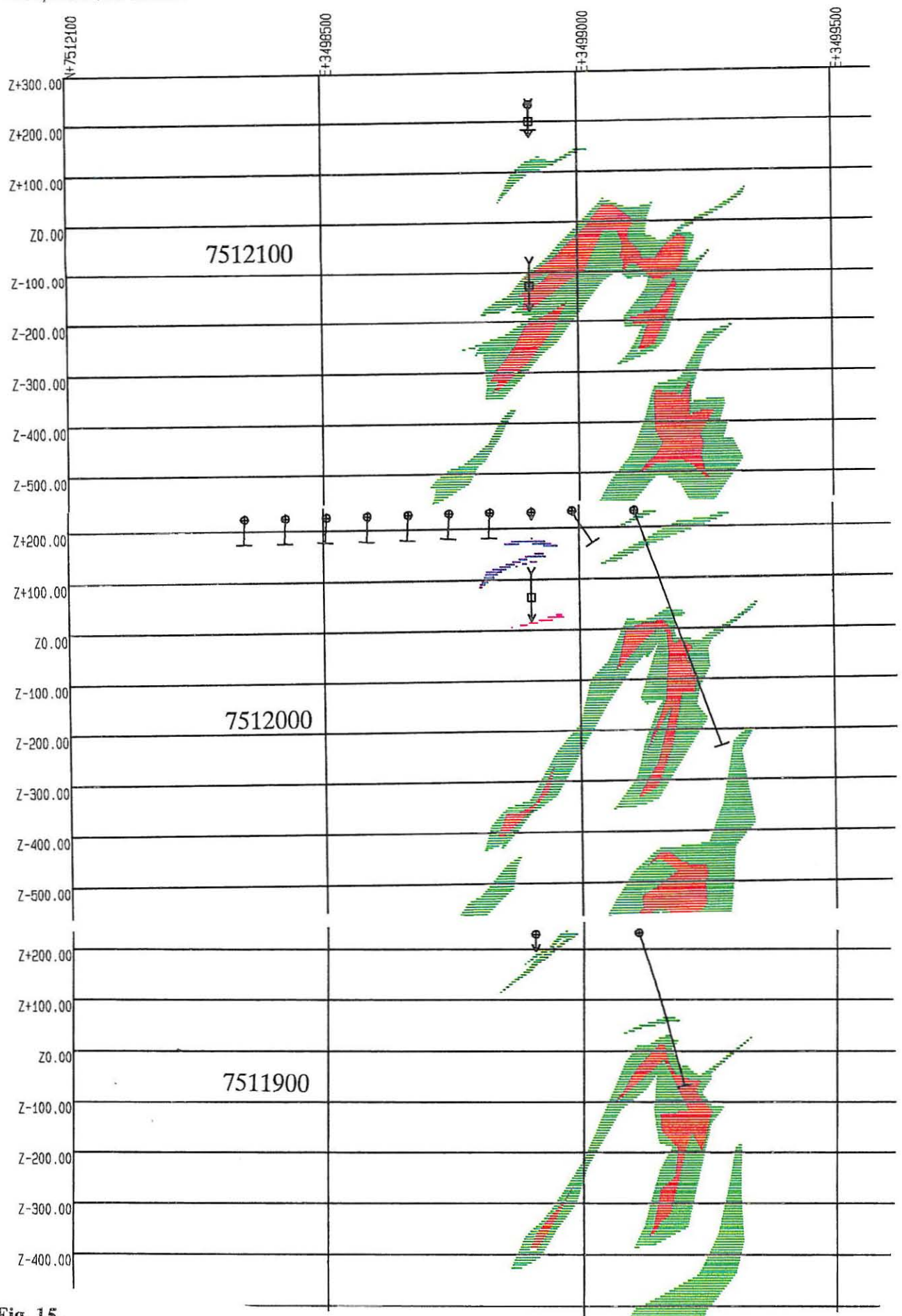


Fig. 15.

Keivitsa, east-west profiles from south to north: 7511900, 7512000, 7512100. Explanation for Ni mineralization as in Fig. 13.

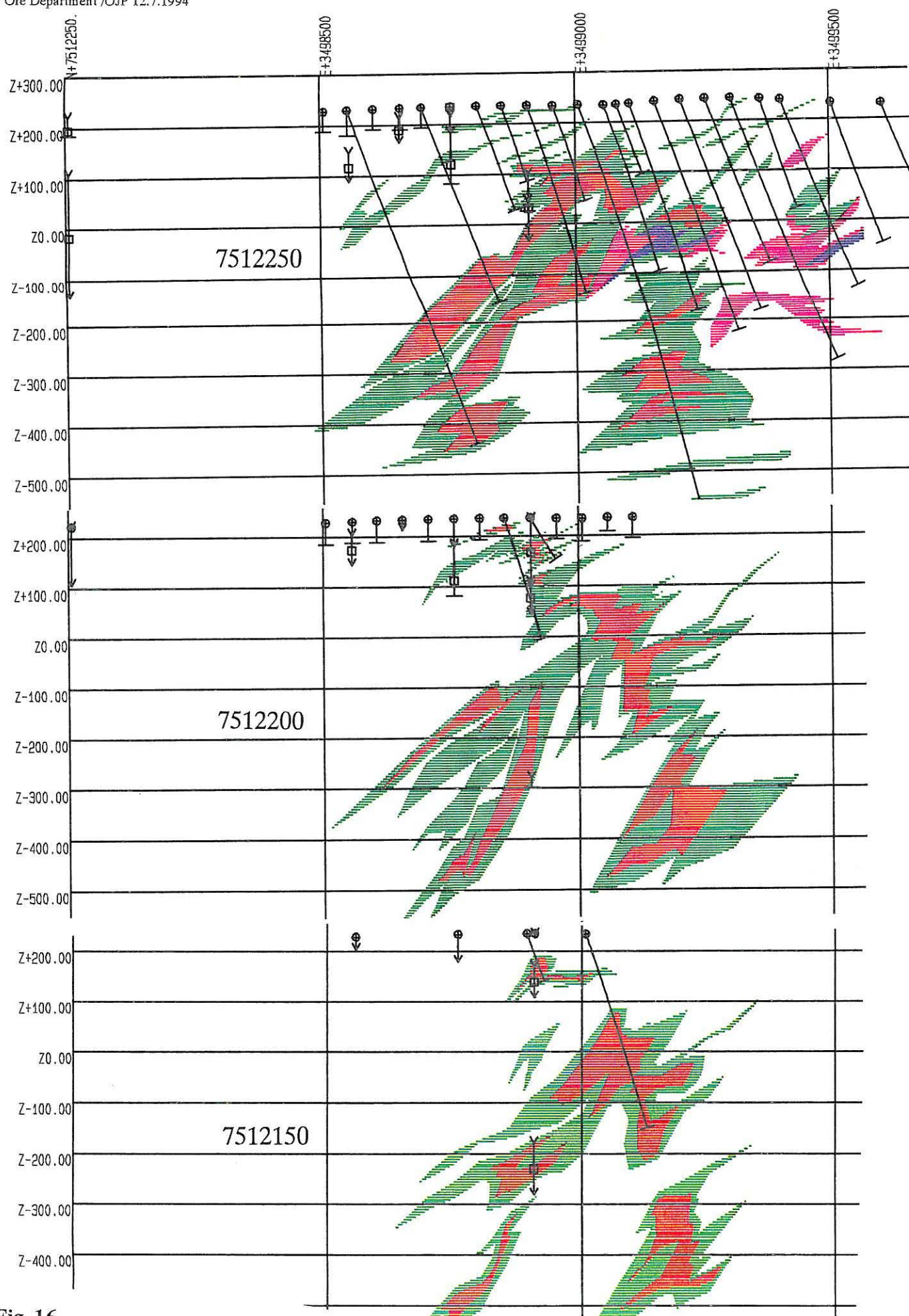


Fig. 16.

Keivitsa, east-west profiles from south to north: 7512150, 7512200, 7512250. Explanation as in Fig. 15.

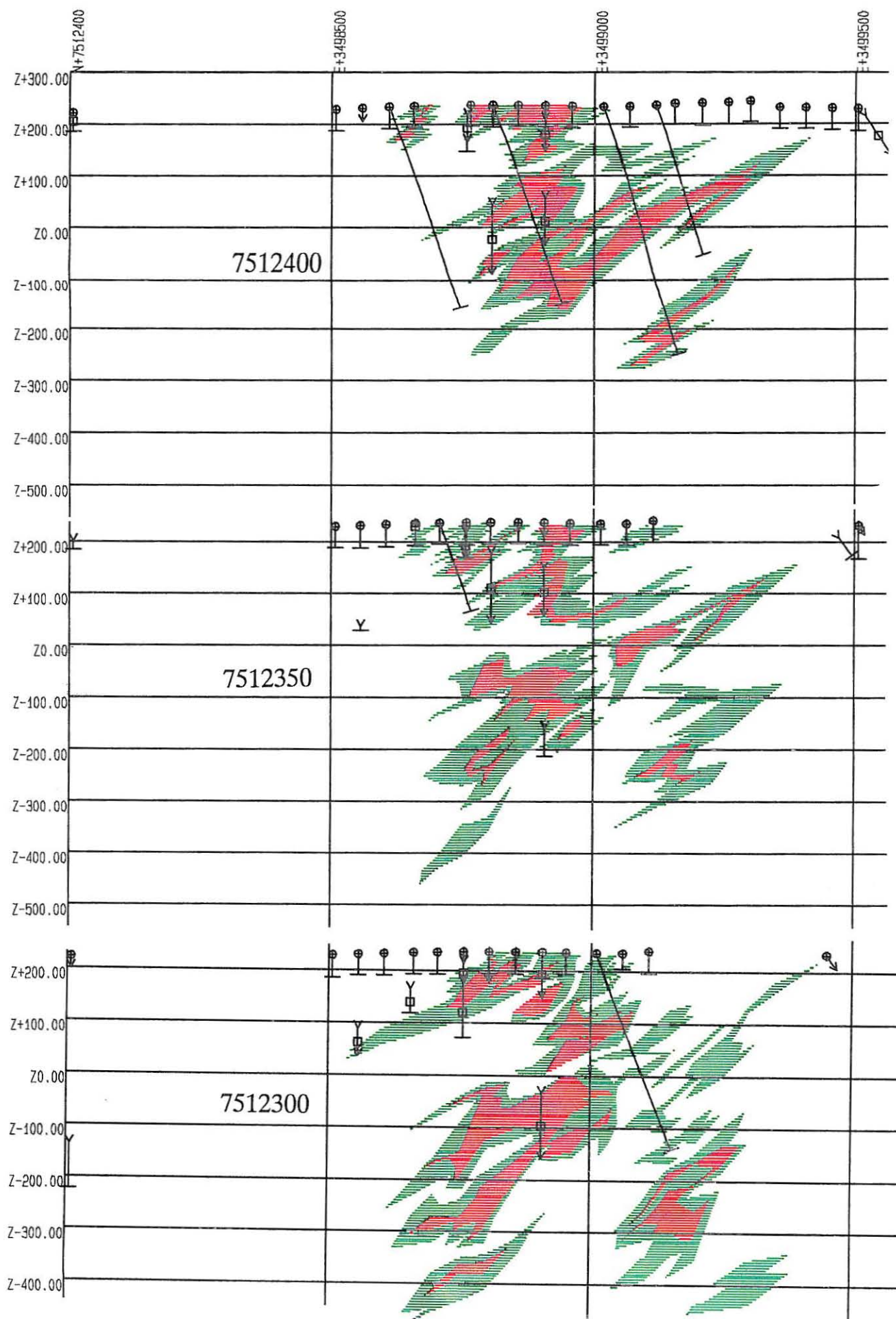


Fig. 17.

Keivitsa, east-west profiles from south to north: 7512300, 7512350, 7512400. Explanation as in Fig.15.

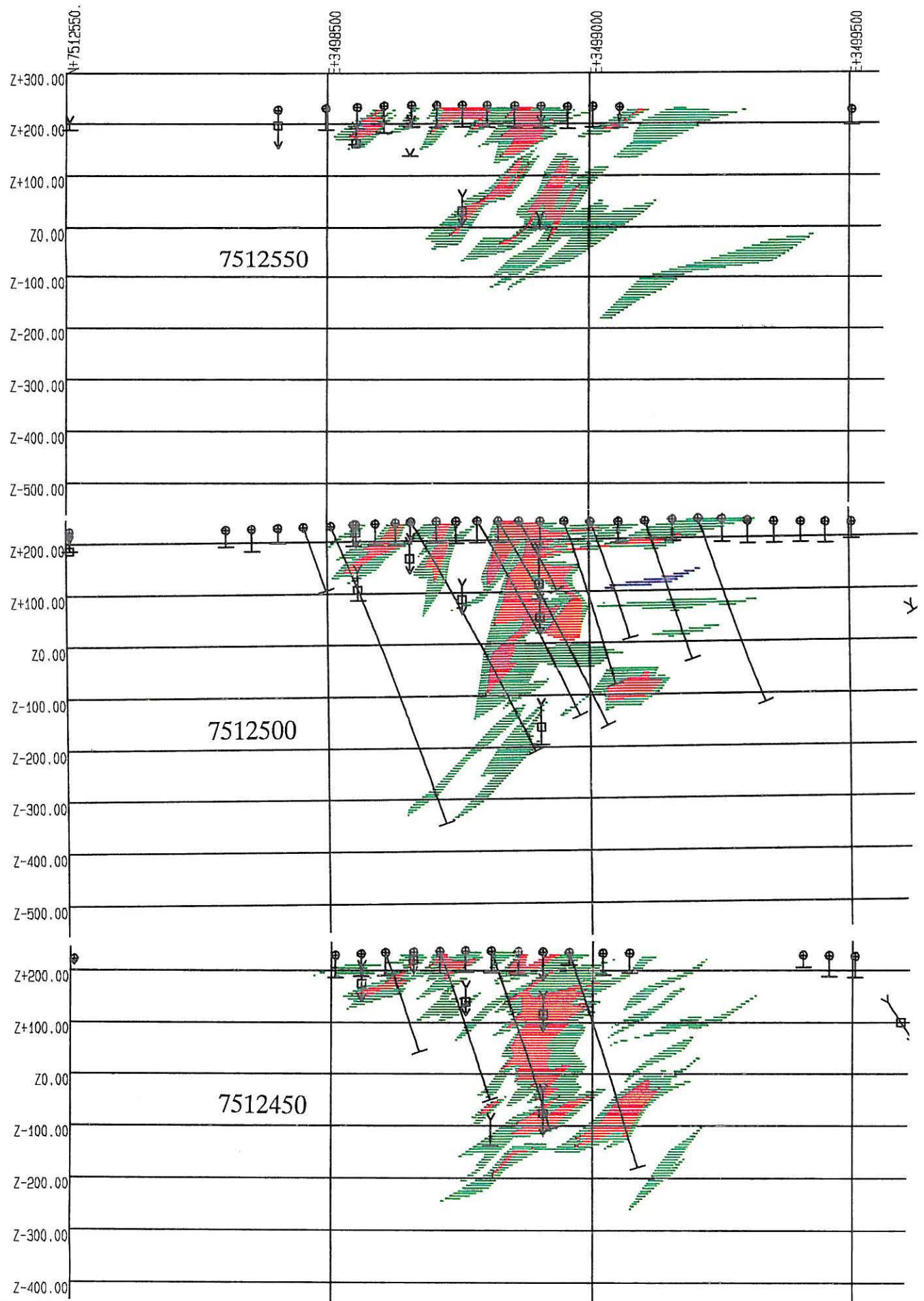


Fig. 18.

Keivitsa, east-west profiles from south to north: 7512450, 7512500, 7512550. Explanation as in Fig. 15.

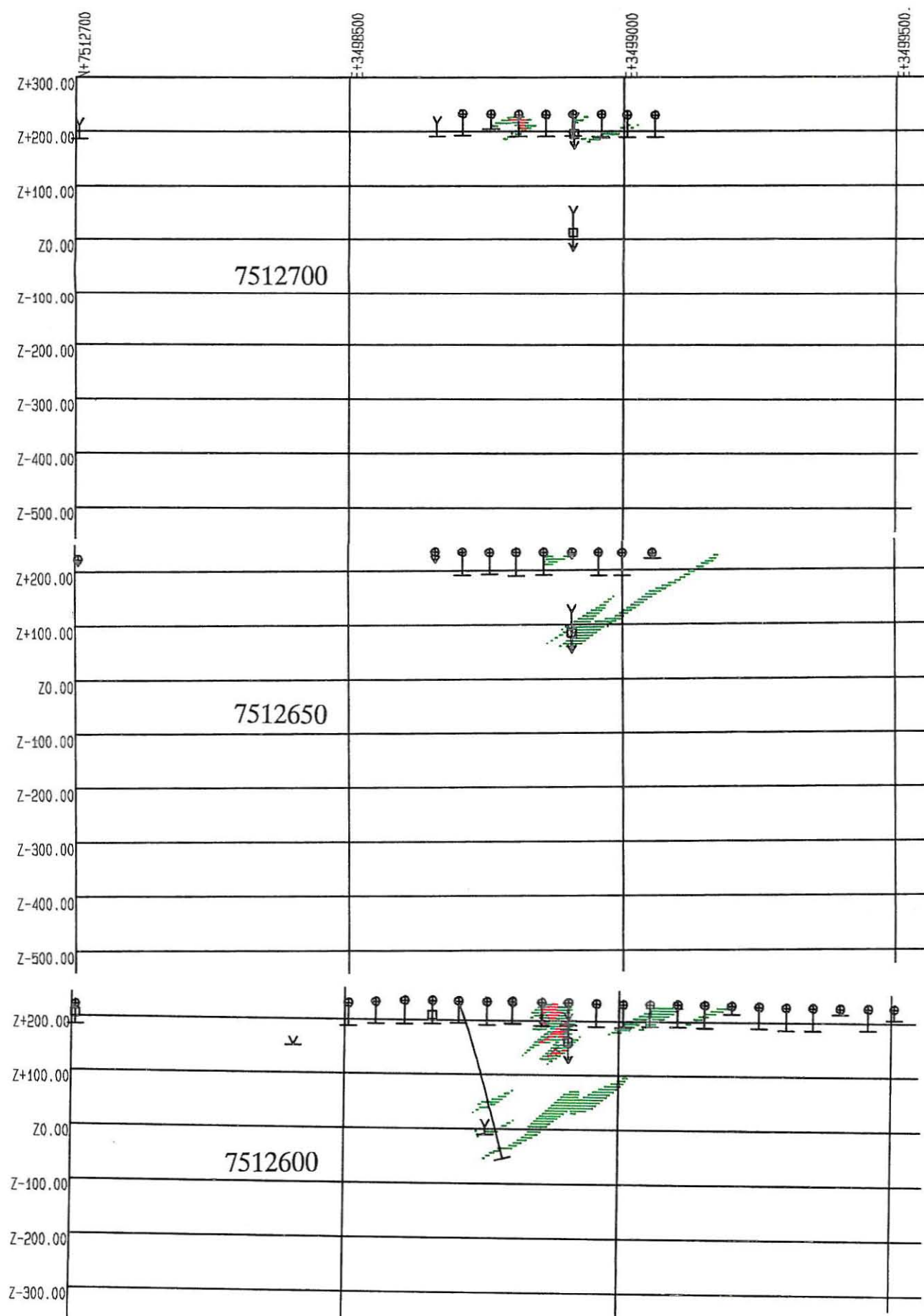


Fig. 19.

Keivitsa, east-west profiles from south to north: 7512600, 7512650, 7512700. Explanation as in Fig. 15.

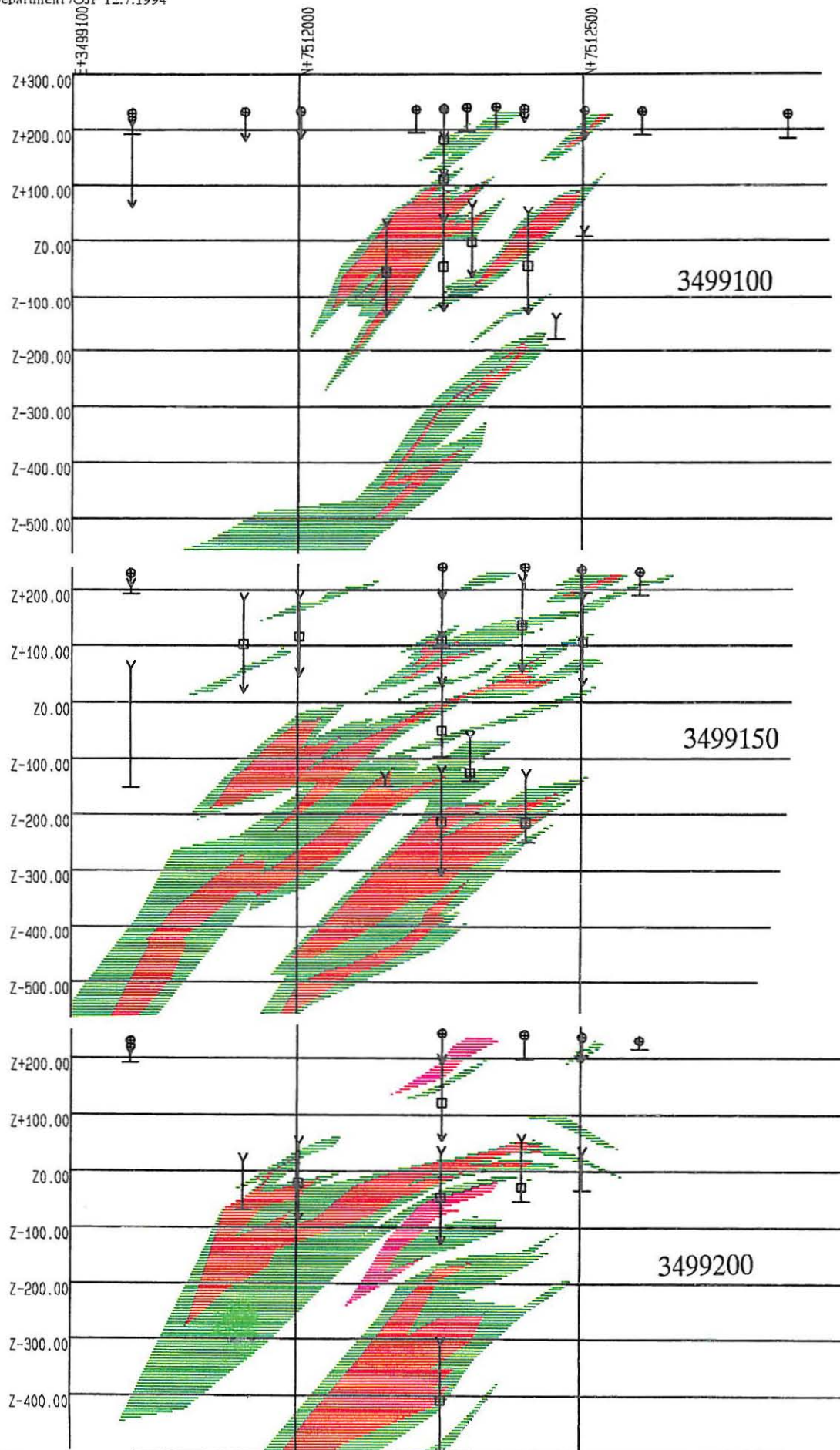


Fig. 20.
Keivitsa, south-north profiles from east to west: 3499200, 3499150, 3499100. Explanation as in Fig. 15.

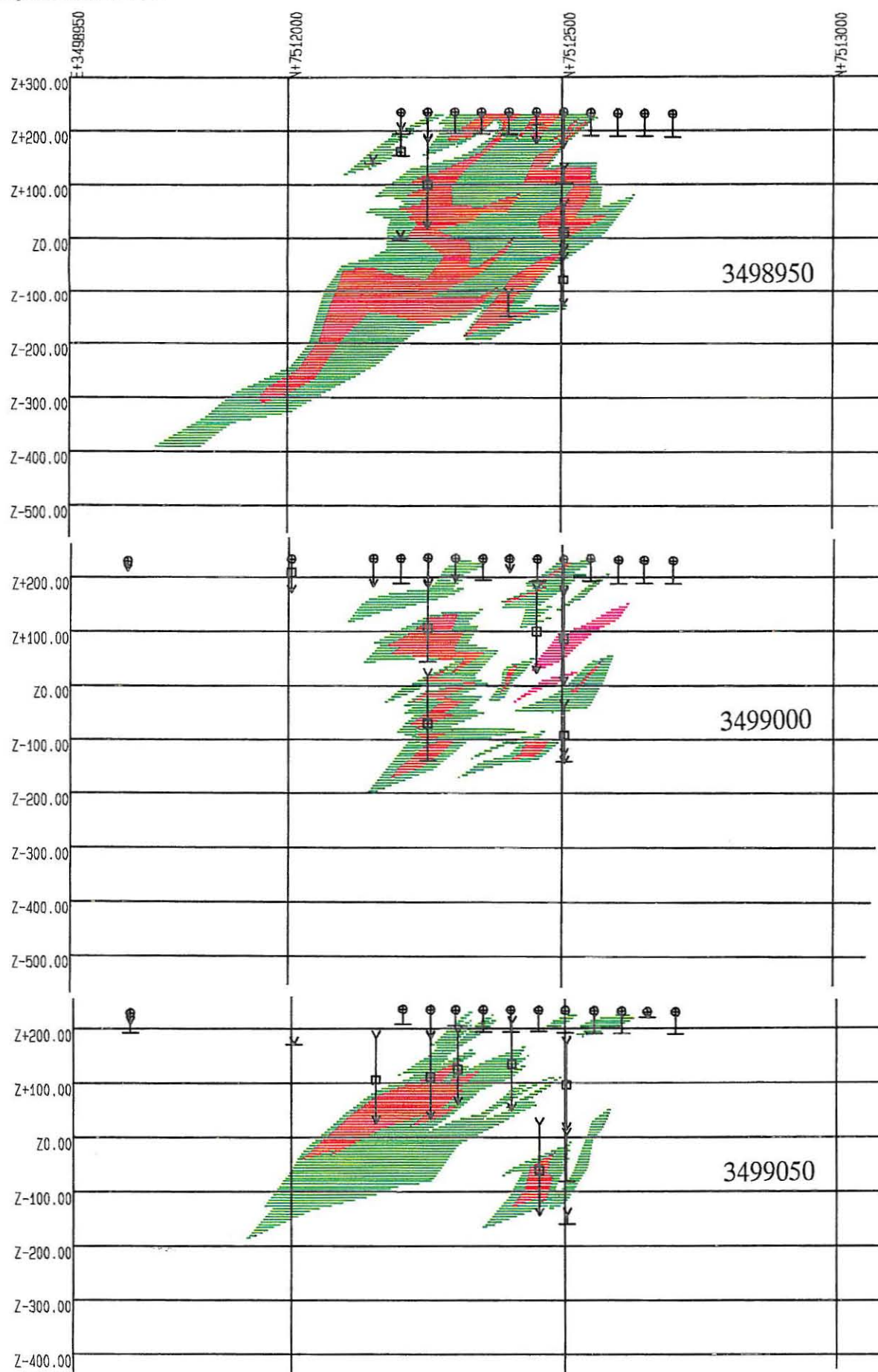


Fig. 21.

Keivitsa, south-north profiles from east to west: 3499050, 3499000, 3498950. Explanation as in Fig. 15.

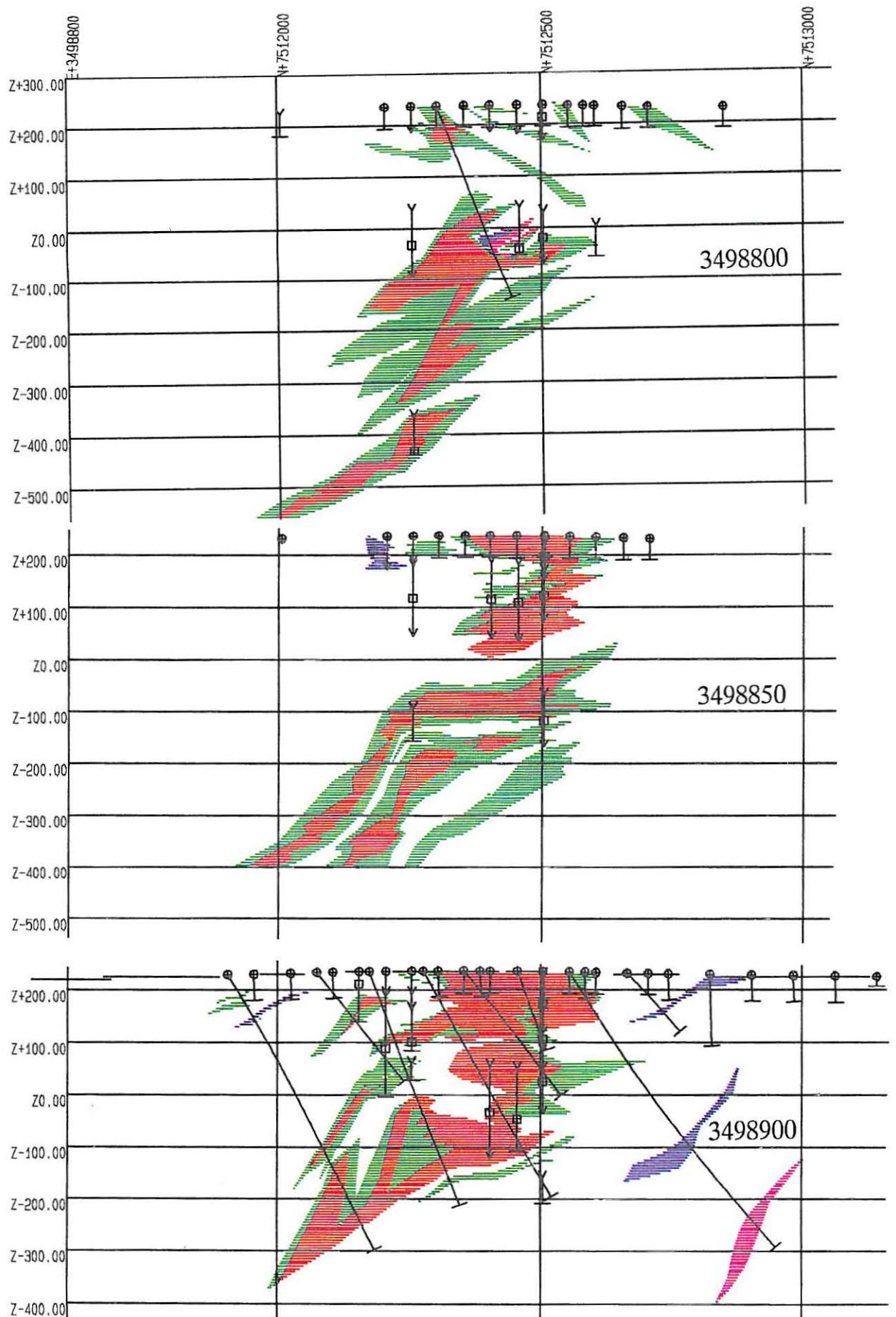


Fig. 22.
Keivitsa, south-north profiles from east to west: 3498900, 3498850, 3498800. Explanation as in Fig. 15.

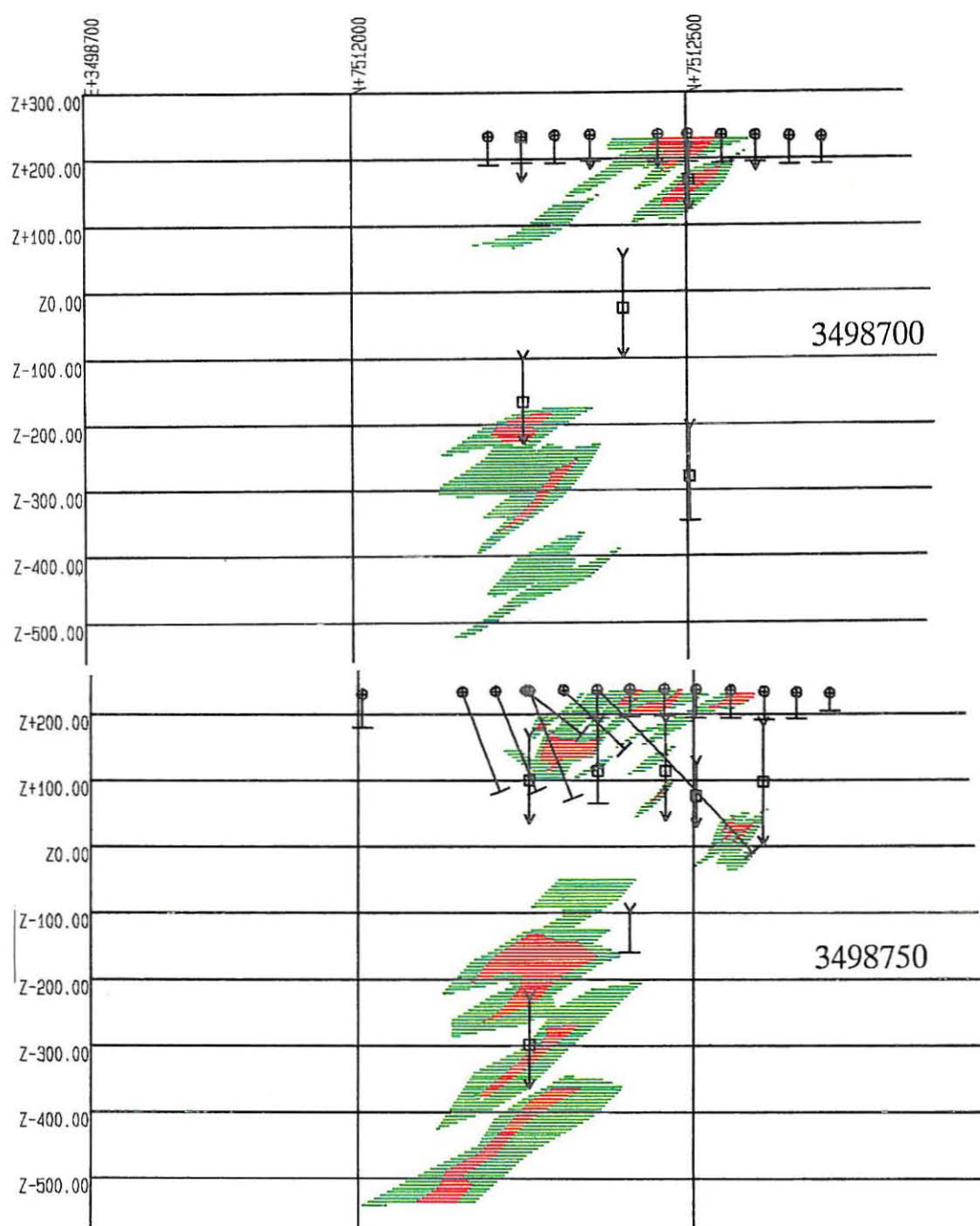


Fig. 23.

Keivitsa, south-north profiles from east to west: 3498750, 3498700. Explanation as in Fig. 15.

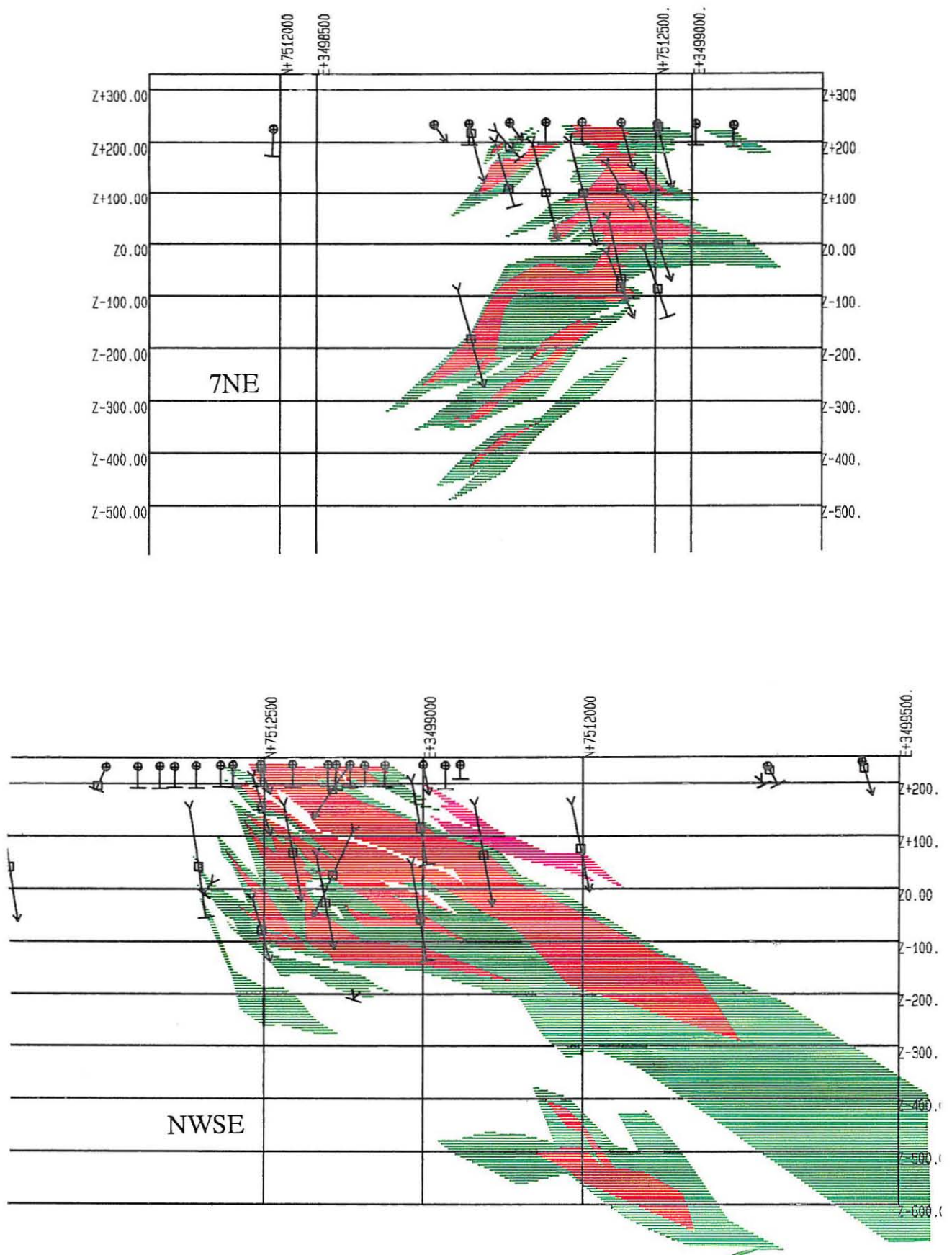


Fig. 24.

Keivitsa, diagonal profiles NWSE and 7NE (SW-NE). Explanation as in Fig. 15.

Tentative mineral resource estimate for the Keivitsa Deposit

The estimate is made in three parts and is based on geological modelling and mineralization outlines on east-west and south-north vertical sections, and on horizontal planes. Vertical sections have been given a thickness of 50 m, which means that the distance of influence of polygons outlined on sections is 25 m in both directions. The corresponding intervals between horizontal planes are 10 m from + 230 level down to + 40 level and 50 m from there down to - 400 level. Tonnage calculations are based on these volumes and on average densities calculated for rock samples of different grade classes and obtained from drill core determinations. Grade averages are arithmetic despite the nearly lognormal appearance of element distributions.

Average grades have been calculated in three ways:

- * **First, *non-weighted averages* for nickel, copper, cobalt, sulphur, platinum, gold, and rhenium, corresponding to different grade classes of each element, were calculated (Table 2.).** Non-weighted averages are also seen in Tables 5 and 7.
- * **Second, *averages weighted with sample lengths*, and corresponding to different cutoff grades of nickel and copper, were calculated for the same elements except rhenium (Table 3.).**
- * **Third, *samples were composited within horizontal planes*, and for these composite samples weighted averages for nickel, copper, cobalt and sulphur, corresponding to different cutoff grades of nickel and copper, were calculated (Table 4.).** Lengths of these composite samples vary from 10 to 14 m depending on the drill hole angle. Each grade is focused to the point where the drill hole in question penetrates the plane involved. These composite samples include original samples without any qualitative filtering. Therefore, **they include diluting samples** in a manner analogous to mined blocks with internal and marginal gangue. Therefore **average grades of composite samples have been used in all three mineral resource estimates in this report (Tables 1, 6).** The only difference is that averages of all classified composite samples were used in vertical section estimates (Table 6A, B), but the averages used in horizontal section estimates were sub-averages of each depth zone (Table 6C).

Besides grade classification, categories recommended by international agencies and collected and refined by The Finnish Association of Mining and Metallurgical Engineers have been adopted. According to the recommendations, even after having located and outlined a potentially economic deposit, it is still not feasible to evaluate a deposit economically before inventory drilling, geotechnical investigations, and pilot plant tests for processing and technical studies of different rock types have been carried out. In summary: **by these criteria, no proven or measured ore reserves are available, but there may be probable and possible (or: indicated and inferred) mineral resources.**

If the geological model described in this report is approximately correct, it seems safe to estimate that the probable resources cover two-thirds of the total resources down to a depth of 300 m (level -70). The marginal parts and the steepest portion (-70 to -400 level) of the deposit comprise the possible resource.

The three mineral resources estimates are summarized in Table 6A-B. The reasons for the different results have been discussed in connection with geological modelling. In the cross section estimates, no distinction between probable and possible resources was made nor any subdivision into depth zones. **In this report, the planar estimate is considered the most accurate and reliable of these three.**

Table 6.

Tentative linear sectional mineral resources estimate for the Keivitsa Deposit done in three ways:

A. Based on south-north cross sections 3498799 - 3499200, section intervals 50 m;

B. Based on east-west sections 7511900 - 7512700, section intervals 50 m;

C. Based on horizontal planes from +230 level to +40 level at 10 m intervals and from +40 level to -400 level at 50 m intervals. **D.** Comparison of three estimates.

Profiles	Cutoff	Density	Volume	Tonnage	Ni	Cu	Au	Pt
Thickness 50 m	% Ni		1000m	Mt	%	%	g/t	g/t
A.								
3498700-3499200	0.1-0.2	3.15	45133	142	0.14	0.17	0.06	0.15
	0.2-max	3.2	29017	92.9	0.32	0.44	0.14	0.35
	0.1-max	3.17	74150	235	0.21	0.28	0.09	0.23
B.								
7511900-7512700	0.1-0.2	3.15	56945	179	0.14	0.17	0.06	0.15
	0.2-max	3.2	26098	83.5	0.32	0.44	0.14	0.35
	0.1-max	3.166	83043	262.5	0.2	0.26	0.09	0.22
C.								
Levels/Depth zones	Cutoff	Density	Probable	Possible	Ni	Cu	Au	Pt
Thickness 10 m	% Ni		Tonnage	Tonnage	%	%	g/t	g/t
			Mt	Mt				
+230 -- +140 / 0 - 100 m	0.1-0.2	3.15	18	2	0.14	0.16	0.06	0.16
	0.2-max	3.2	8.6	1	0.32	0.36	0.13	0.47
+230 -- +40 / 0 - 200 m	0.1-0.2	3.15	32	10	0.14	0.17	0.07	0.16
	0.2-max	3.2	16	4	0.32	0.43	0.15	0.42
+230 -- -65 / 0 - 300 m	0.1-0.2	3.15	52	25	0.14	0.17	0.06	0.16
	0.2-max	3.2	24	11	0.32	0.44	0.15	0.4
-60 -- -400/ 300 - 630 m	0.1-0.2	3.15		116	0.14	0.17	0.06	0.15
	0.2-max	3.2		36	0.31	0.43	0.13	0.23
Sum total :								
+230 -- -400 /	0.1-0.2	3.15	52	141	0.14	0.17	0.06	0.15
0 - 630 m	0.2-max	3.2	24	47	0.32	0.44	0.13	0.35
D.								
Comparison of total resources:								
Sections	Cutoff	Density		Tonnage	Ni	Cu	Au	Pt
	% Ni			Mt	%	%	g/t	g/t
3498700-3499200	0.1-max	3.17		235	0.21	0.28	0.09	0.23
7511900-7512700	0.1-max	3.166		262.5	0.2	0.26	0.09	0.22
+230 -- -400	0.1-max	3.16		264	0.19	0.24	0.08	0.2

Recommendations for further prospecting

The Keivitsa Deposit has been tentatively outlined. The next phase of exploration should attempt to give it the status of an ore deposit. A necessary part of further studies will be geological research including structural, petrographical, mineralogical, geochemical and geophysical approaches and followed by mine technical and economical studies. A few aspects from a geologist's point of view are listed below.

- * Many potentially valuable minerals have been identified; their behaviour separately and together should be examined.
- * The internal structure of the Keivitsa intrusive, and the position of the mineralization in the structure, or, its structural control should be examined. This requires cooperation between geologists and geophysicists and would also necessitate some more drilling into obvious pig holes or into predicted but uncertain extensions of the deposit (see Figs. 15 - 17).
- * The physical and chemical features of Keivitsa's rock and mineralization types should be examined to anticipate their behaviour in mining and refinement processes.
- * The Keivitsa Deposit may not be unique or the best representative of its kind. Again, it involves cooperation of geologists, geophysicists and geochemists to create a deposit model for forward exploration.
- * Only a combination of multifaceted investigations will produce the parameters necessary to find out the best equivalence choices and optimum production patterns for the evaluation of the Keivitsa Deposit.

Reference

Vuorimiesyhdistys-Bergsmannaföreningen ry. , 1991, Malmiarvion laatiminen, 111 p., Sarja A No 97.(The Finnish Association of Mining and Metallurgical Engineers, Guide to ore reserve estimates).

Appendix: Noble element statistics of combined samples.

Assay unit is ppb except with Ni, Cu, S.

Table 7.

A. Statistics of 3984 assays of combined noble metal samples, and corresponding Ni, Cu and S assay averages.

	Au	Ir	Os	Pd	Pt	Re	Rh	Ni%	CU%	S%
Data Min.	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0	0	0.01
Data Max.	2070	92.3	140	4270	5000	59	312	1.77	7.79	23.4
Mean	73.87154	4.9943	4.40505	118.19429	187.64665	0.67234	7.61157	0.17523	0.21286	1.03318
Geom. Mean	29.71535	1.53671	0.56043	48.59426	84.70981	0.10477	1.64212	0.12856	0.11148	0.46831
Ln. Mean	3.39166	0.42964	-0.57905	3.88351	4.43923	-2.25598	0.49599	-2.05135	-2.19391	-0.75862
Variance	10158.67	49.63259	46.19638	39073.978	89846.484	2.95317	182.1485	0.02255	0.07311	1.4062
Ln Variance	3.24423	4.58574	8.99857	2.83846	2.18201	5.11464	5.97692	0.65172	1.59528	2.78544
Stand. Dev.	100.7902	7.04504	6.79679	197.67139	299.74403	1.71848	13.49624	0.15016	0.27038	1.18583
C. of Var.	1.3644	1.41061	1.54295	1.67243	1.59739	2.55595	1.77312	0.85692	1.27026	1.14775

B. Correlation matrix of 3984 assays of combined noble metal samples and corresponding Ni, Cu, and S assay averages.

	Au	Ir	Os	Pd	Pt	Re	Rh	Ni%	CU%	S%
Au	-----	0.35829	0.31569	0.38231	0.37793	0.03873	0.39995	0.5259	0.54588	0.37242
Ir	0.35829	-----	0.83489	0.71316	0.79444	0.05984	0.66011	0.64238	0.24424	0.20449
Os	0.31569	0.83489	-----	0.61829	0.69903	0.05902	0.54814	0.54892	0.21842	0.1987
Pd	0.38231	0.71316	0.61829	-----	0.92938	0.01685	0.48977	0.67547	0.21047	0.1788
Pt	0.37793	0.79444	0.69903	0.92938	-----	0.00812	0.54569	0.65412	0.22221	0.16791
Re	0.03873	0.05984	0.05902	0.01685	0.00812	-----	0.11576	0.14533	0.10691	0.44063
Rh	0.39995	0.66011	0.54814	0.48977	0.54569	0.11576	-----	0.55884	0.49999	0.32308
Ni%	0.5259	0.64238	0.54892	0.67547	0.65412	0.14533	0.55884	-----	0.58111	0.46731
CU%	0.54588	0.24424	0.21842	0.21047	0.22221	0.10691	0.49999	0.58111	-----	0.46328
S%	0.37242	0.20449	0.1987	0.1788	0.16791	0.44063	0.32308	0.46731	0.46328	-----